

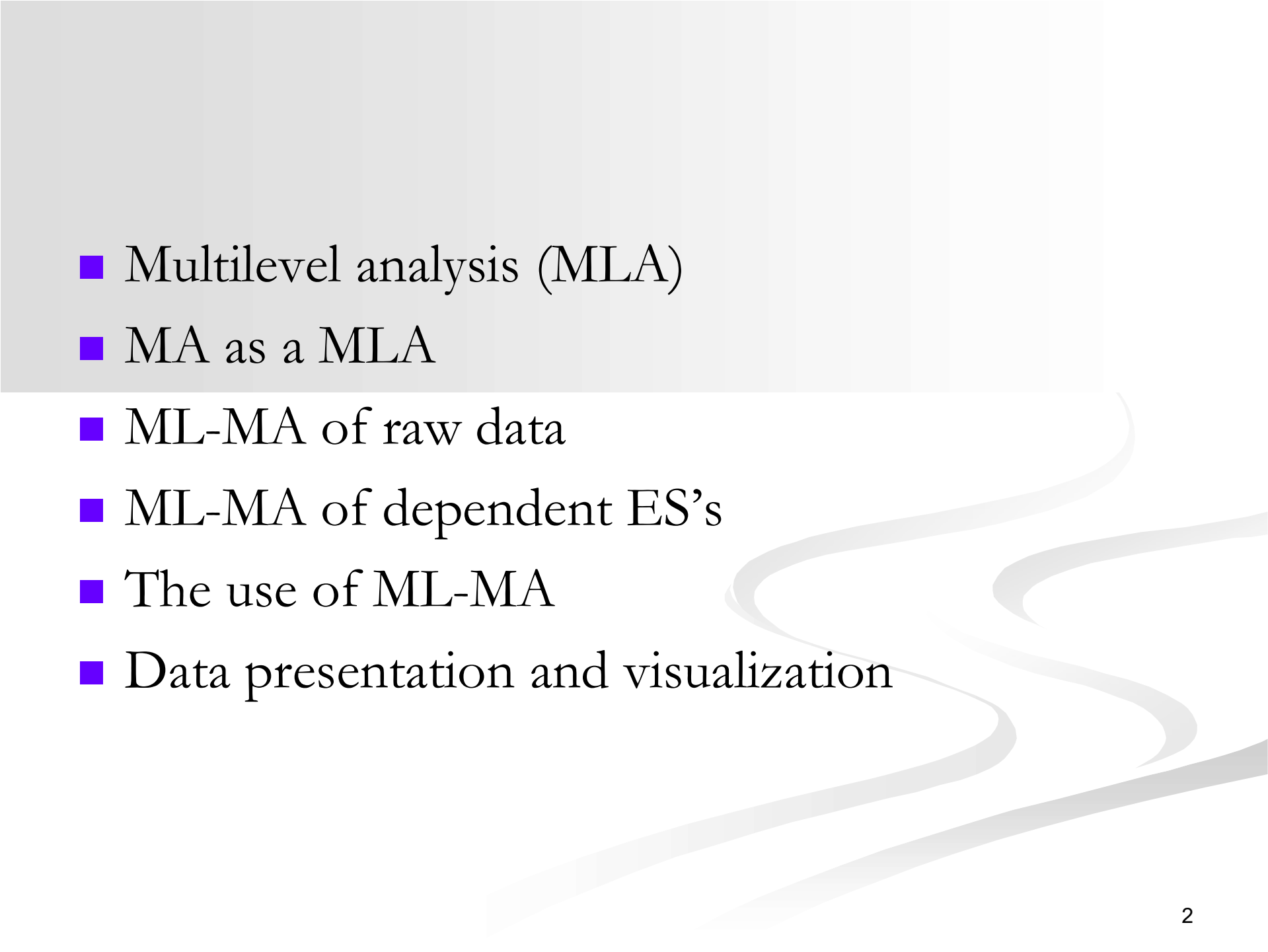
Using multilevel models for the meta-analysis of dependent effect sizes

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- 
- Multilevel analysis (MLA)
 - MA as a MLA
 - ML-MA of raw data
 - ML-MA of dependent ES's
 - The use of ML-MA
 - Data presentation and visualization

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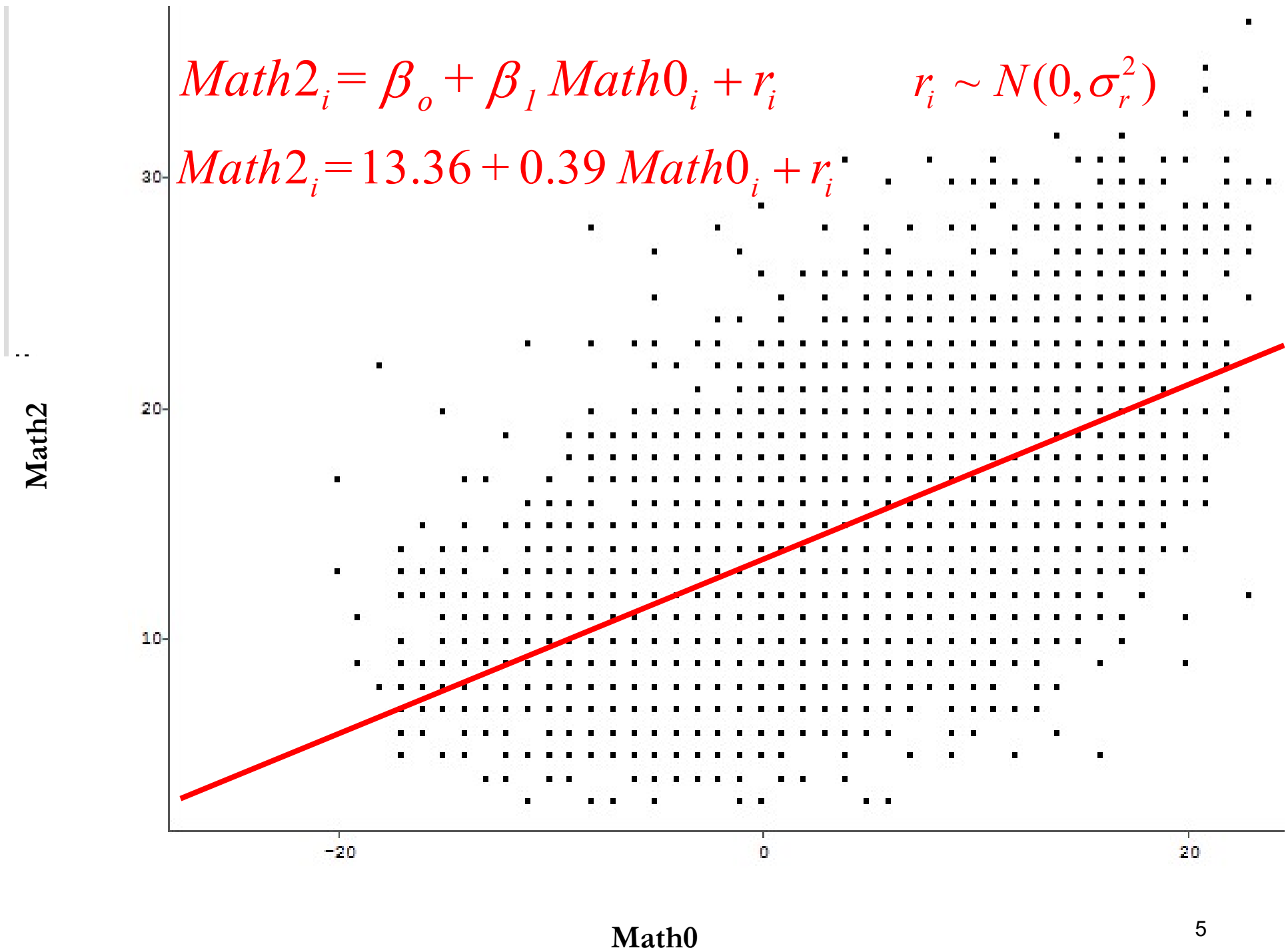
Example: longitudinal research in secondary education in Flanders

6411 students, 57 schools

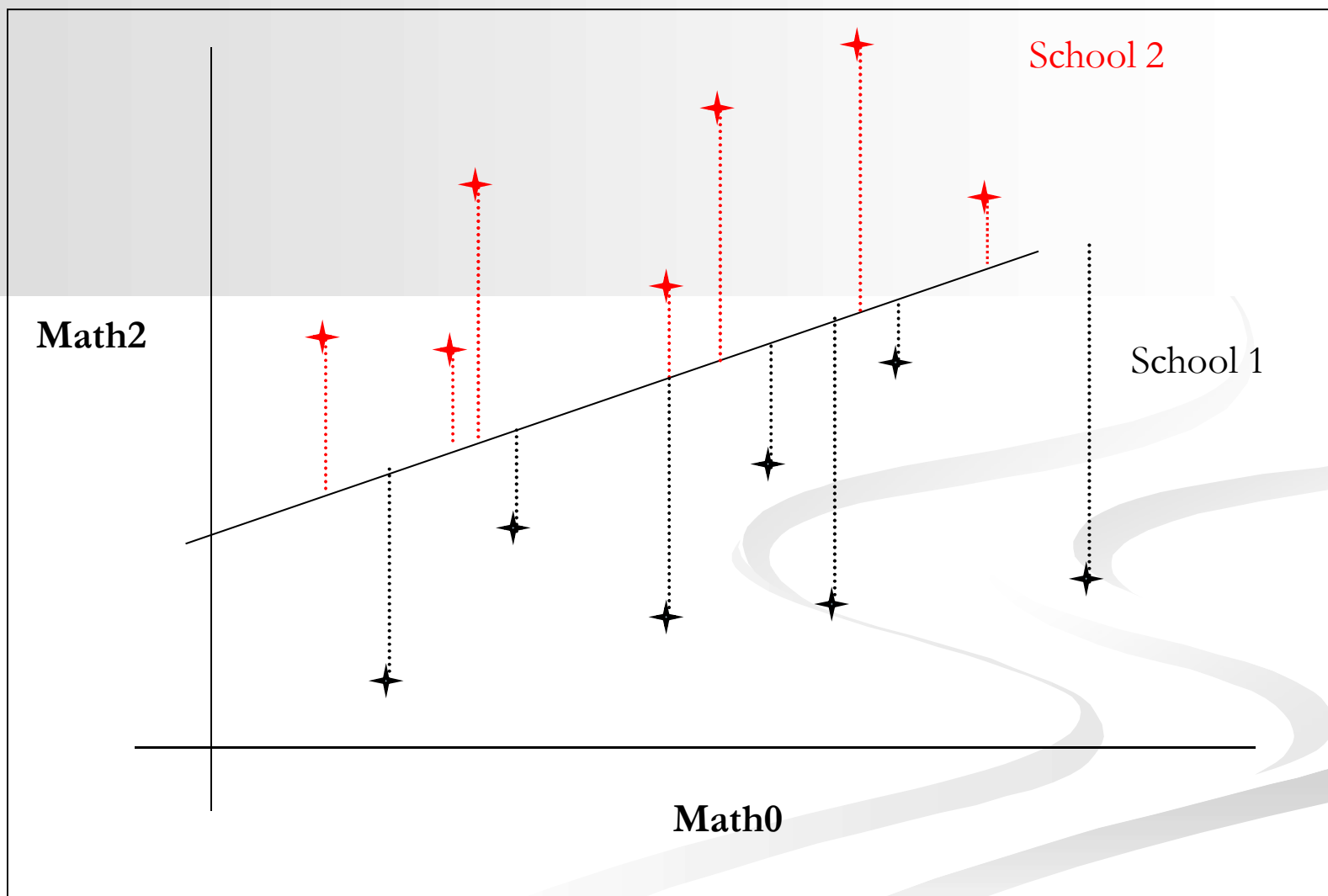
5 measurement moments

1990-1996

Van Damme, J., De Fraine, B., Van Landeghem, G., Opdenakker, M.-C., & Onghena, P. (2002). A new study on educational effectiveness in secondary schools in Flanders: An introduction. *School Effectiveness and School Improvement*, 13, 383-397.



Students nested in schools => Residuals may not be independent!



Problem of ignoring school?

- Overestimation of information

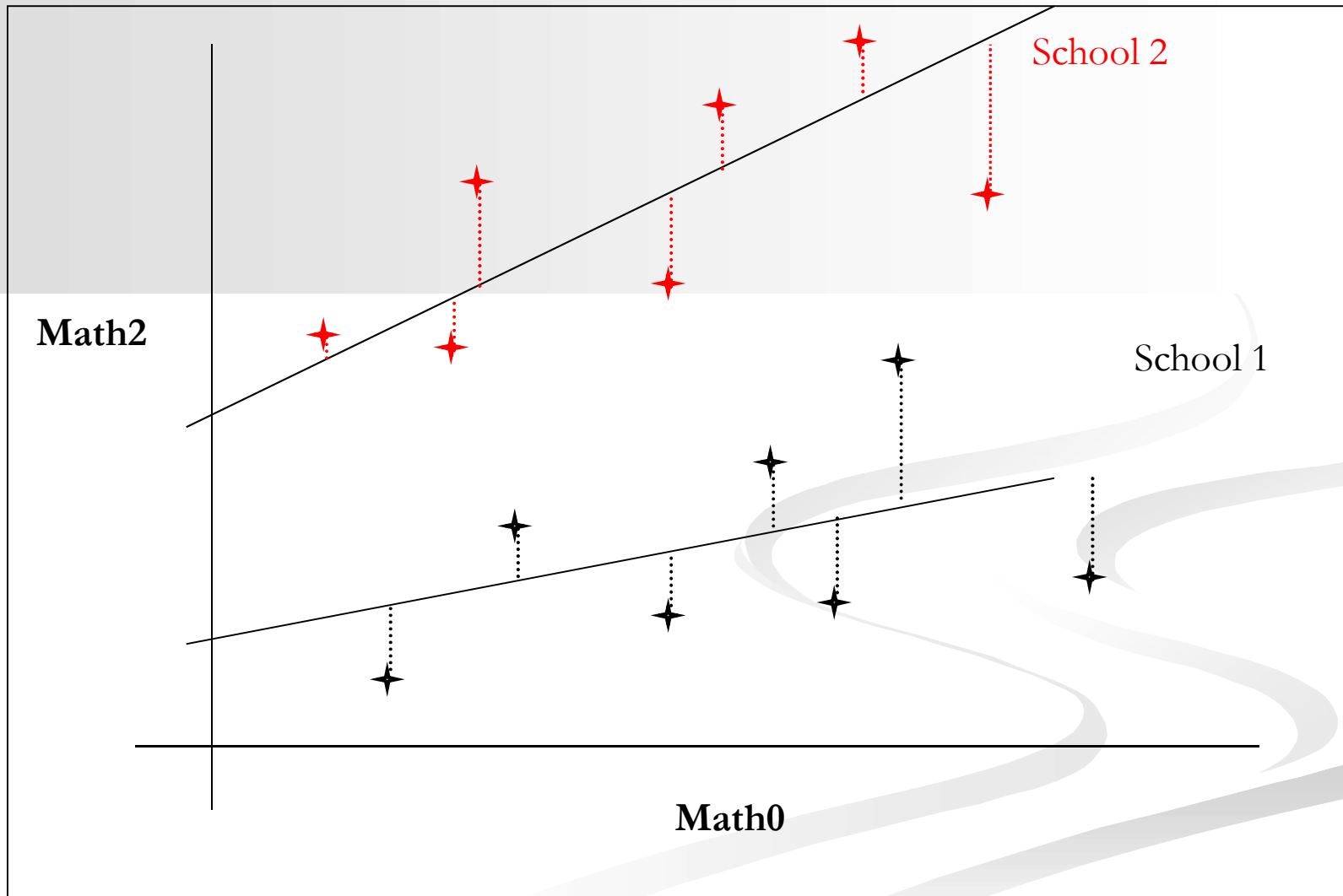
Result:

Standard errors ↓



Type I errors ↑

Solution?



Solution?

Separate regression equations?

$$Y_{i1} = \beta_{01} + \beta_{11}X_{i1} + r_{i1}$$

$\vdots$

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + r_{ij}$$

$\vdots$

$$Y_{iJ} = \beta_{0J} + \beta_{1J}X_{iJ} + r_{iJ}$$

Solution?

School as predictor?

$$Y_i = \beta_o + \beta_1 SCHOOL2_i + \dots +$$
$$(\beta_3 + \beta_4 SCHOOL2_i + \dots) * X_i$$
$$+ r_i$$

Solution: MLA!

(Raudenbush, Goldstein)

Level 1: Student or within school level

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + r_{ij} \quad r_{ij} \sim N(0, \sigma_r^2)$$

Level 2: (Between) school level

$$\begin{aligned}\beta_{0j} &= \gamma_{00} + u_{0j} \\ \beta_{1j} &= \gamma_{10} + u_{1j}\end{aligned}$$

$$\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} \sim N(0, \Omega_u) \text{ with } \Omega_u = \begin{bmatrix} \sigma_{u_0}^2 & \\ \sigma_{u_0u_1} & \sigma_{u_1}^2 \end{bmatrix}$$

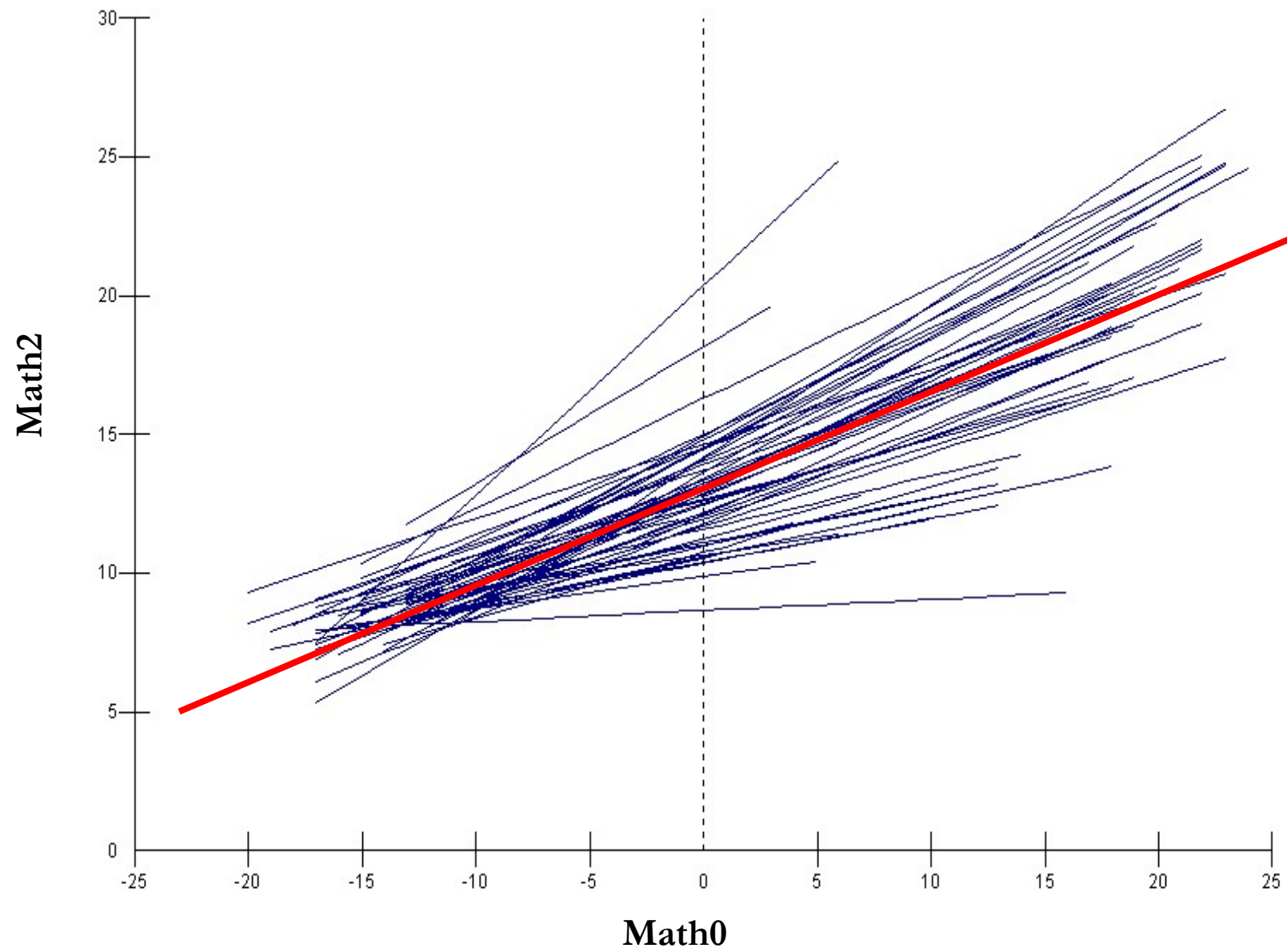
Example

$$Math2_{ij} = \beta_{0j} + \beta_{1j}(Math0)_{ij} + r_{ij} \quad r_{ij} \sim N(0, 15.94)$$

$$\beta_{0j} = 13.22 + u_{0j}$$

$$\beta_{1j} = 0.35 + u_{1j}$$

$$\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} \sim N(0, \Omega_u) \text{ with } \Omega_u = \begin{bmatrix} 4.39 & \\ 0.23 & 0.02 \end{bmatrix}$$



Exploring differences between schools

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + r_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

Exploring differences between schools

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + r_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01}Z_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11}Z_j + u_{1j}$$

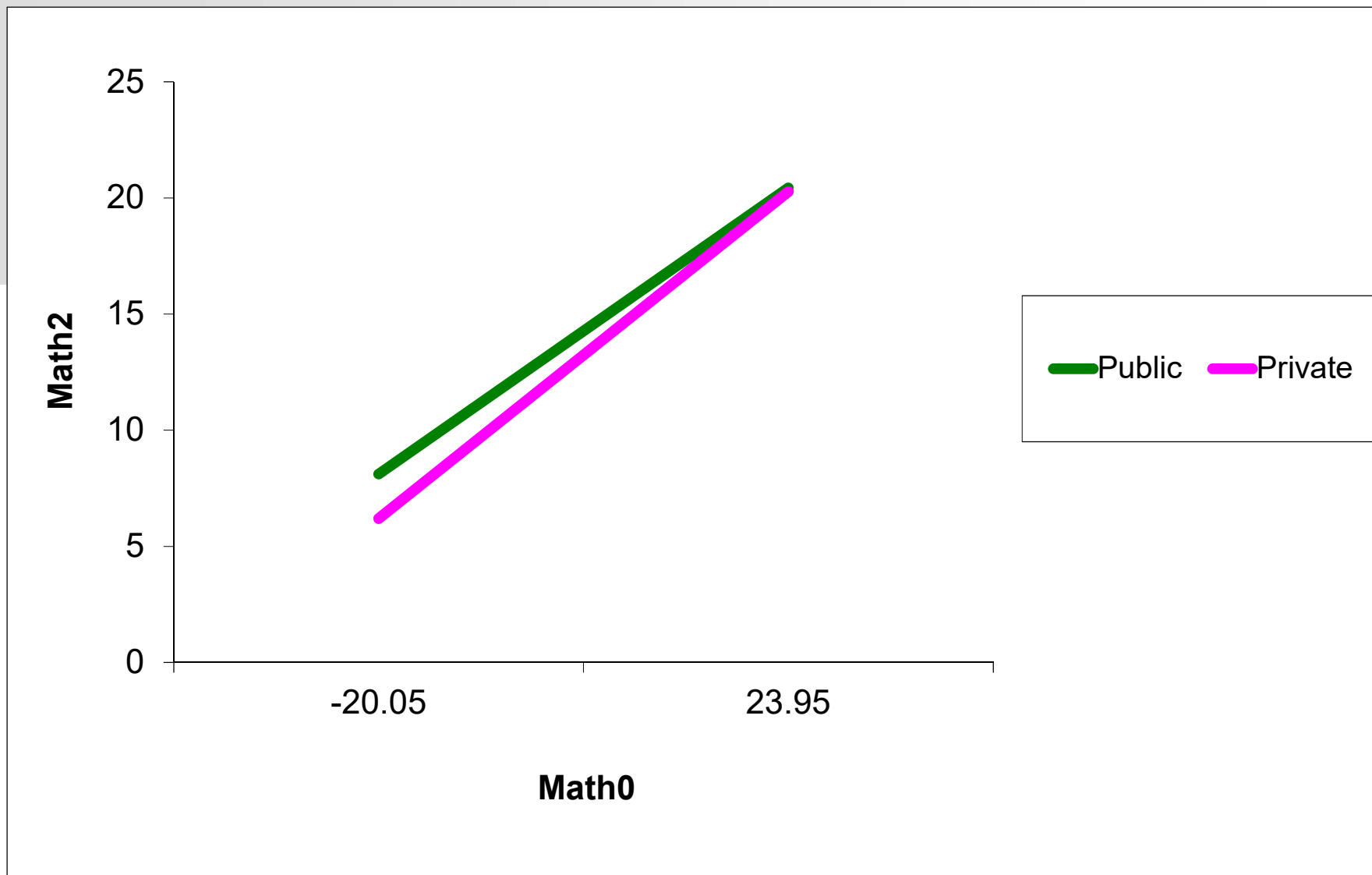
Example

$$Math2_{ij} = \beta_{0j} + \beta_{1j}(Math0)_{ij} + r_{ij} \quad r_{ij} \sim N(0, 15.32)$$

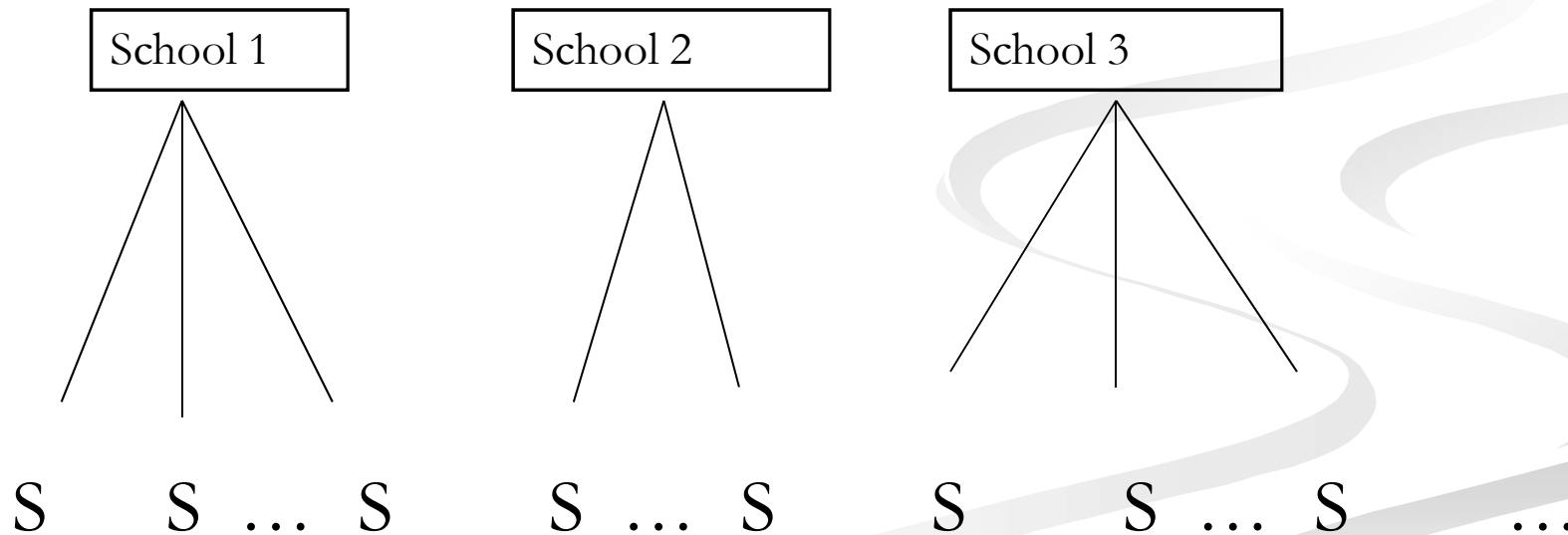
$$\beta_{0j} = 13.73 - 1.12 (PRIVATE)_j + u_{0j}$$

$$\beta_{1j} = 0.28 + 0.04 (PRIVATE)_j + u_{1j}$$

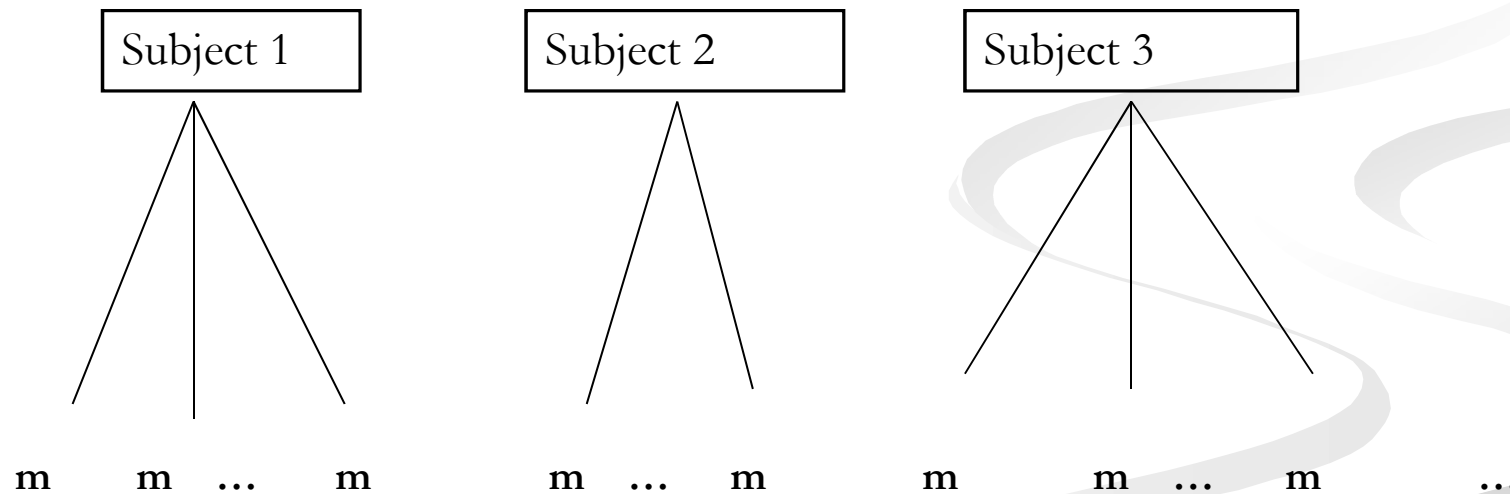
$$\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} \sim N(0, \Omega_u) \quad \text{with } \Omega_u = \begin{bmatrix} 4.09 & \\ 0.24 & 0.02 \end{bmatrix}$$



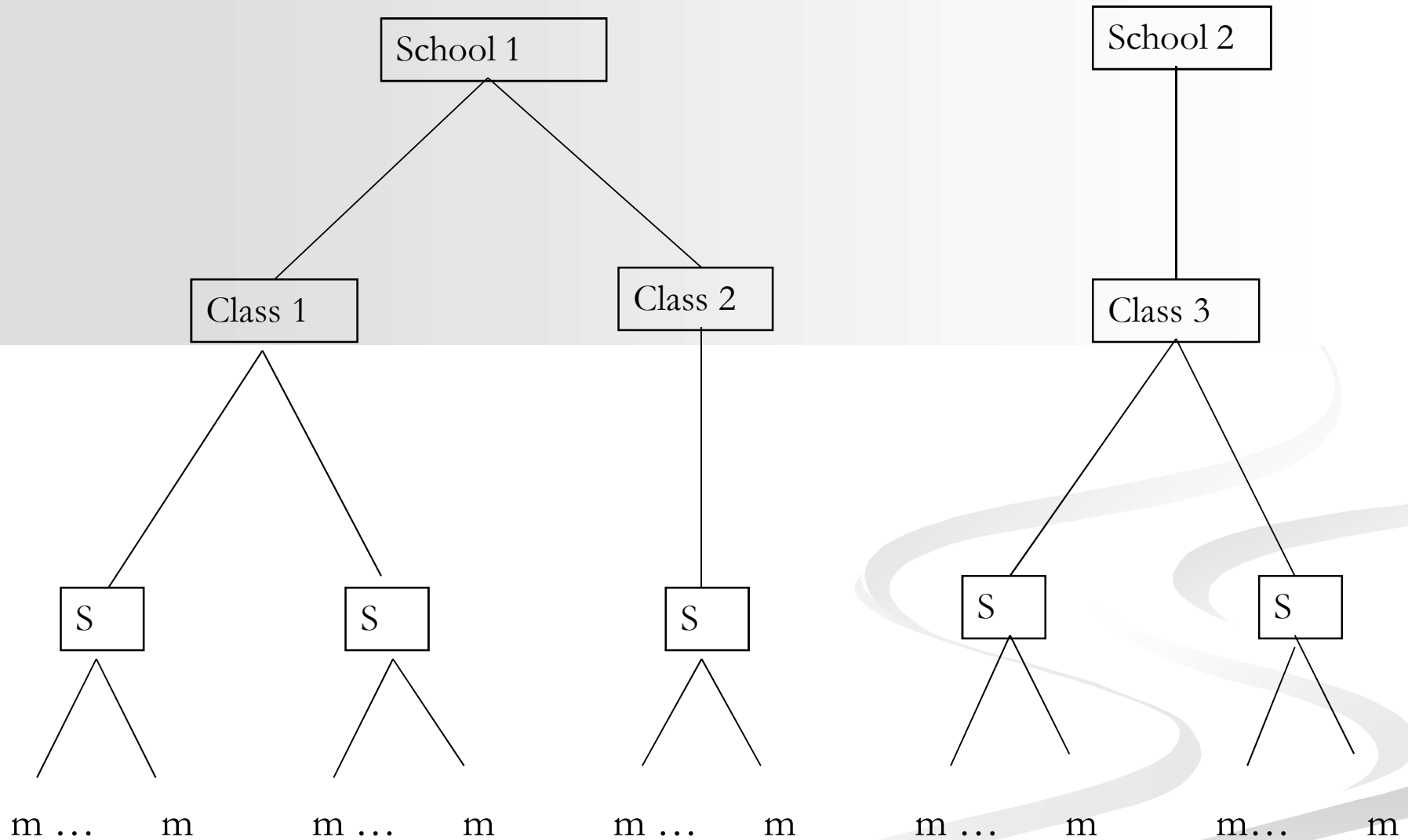
Two-level structure (students nested within schools)



Two-level structure of repeated measurements



Four-level structure



$$Y_{ijk} = \beta_{0jk} + \beta_{1jk}X_{ijk} + r_{ijk} \quad r_{ijk} \sim N(0, \sigma_r^2)$$

$$\begin{cases} \beta_{0jk} = \gamma_{00k} + \gamma_{01k}Z_{jk} + u_{0jk} \\ \beta_{1jk} = \gamma_{10k} + \gamma_{11k}Z_{jk} + u_{1jk} \end{cases} \quad \begin{bmatrix} u_{0jk} \\ u_{1jk} \end{bmatrix} \sim N(0, \Omega_u)$$

$$\begin{cases} \gamma_{00k} = \theta_{000} + \theta_{001}W_k + v_{00k} \\ \gamma_{01k} = \theta_{010} + \theta_{011}W_k + v_{01k} \\ \gamma_{10k} = \theta_{100} + \theta_{101}W_k + v_{10k} \\ \gamma_{11k} = \theta_{110} + \theta_{111}W_k + v_{11k} \end{cases} \quad \begin{bmatrix} v_{00k} \\ v_{01k} \\ v_{10k} \\ v_{11k} \end{bmatrix} \sim N(0, \Omega_v)$$

Why using multilevel models?

- Statistical reason:
 - Accounts for overlap in information, resulting in an appropriate Type I error rate.
- Conceptual reason:
 - Information about differences between students and between schools

- Multilevel analysis (MLA)
- MA as a MLA
- ML-MA of raw data
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- Use of ML-MA
- Data presentation and visualization

Schools

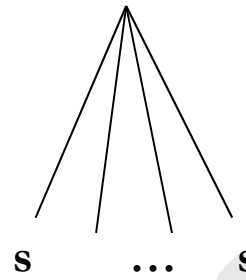
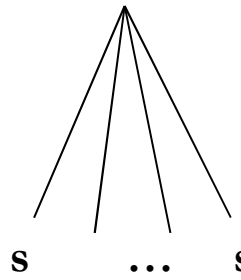
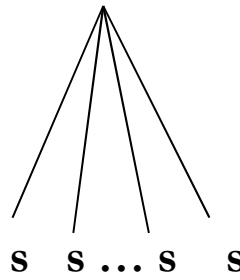
School 1

School 2 ...

School k

- mean?
- variance?
- school characteristics?

Students



- variance?
- student characteristics?

Study

Study 1

Study 2 ...

Study k

- mean?
- variance?
- study characteristics?

Participants

pp pp ... pp

pp pp ... pp

pp pp ... pp

- variance?
- participant characteristics

A ML-MA

$$Y_{ij} = \beta_{0j} + \beta_{1j} \text{Treatment}_{ij} + r_{ij} \quad r_{ij} \sim N(0, \sigma_r^2)$$

μ_{jC} $(\mu_{jE} - \mu_{jC})$

$$\beta_{0j} = \gamma_{00} + u_{0j} \quad \text{Overall effect}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

Heterogeneity

$$\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} \sim N(0, \Omega_u) \text{ with } \Omega_u = \begin{bmatrix} \sigma_{u_0}^2 & \\ \sigma_{u_0 u_1} & \sigma_{u_1}^2 \end{bmatrix}$$

A simulated data set

$$k=30$$

$$n_{jC} = n_{jE} = 30$$

Population model:

$$Y_{ij} = \beta_{0j} + \beta_{1j}Treatment_{ij} + r_{ij} \quad r_{ij} \sim N(0, 1)$$

$$\beta_{0j} = 1.00 + u_{0j} \quad u_{0j} \sim N(0, 0.5)$$

$$\beta_{1j} = 1.00 + u_{1j} \quad u_{1j} \sim N(0, 0.5)$$

Parameter estimates

$$Y_{ij} = \beta_{0j} + \beta_{1j}Treatment_{ij} + r_{ij} \quad r_{ij} \sim N(0, 1.00)$$

$$\beta_{0j} = 0.95 + u_{0j} \quad u_{0j} \sim N(0, 0.40)$$

$$\beta_{1j} = 1.19 + u_{1j} \quad u_{1j} \sim N(0, 0.46)$$

Standardizing data

$$\frac{\mu_{jE} - \mu_{jC}}{\sigma_{r_j}} = \text{SMD!}$$

$$\frac{Y_{ij}}{\sigma_{r_j}} = \beta'_{0j} + \beta'_{1j} \text{Treatment}_{ij} + r'_{ij}$$

$$\beta'_{0j} = \gamma'_{00} + u'_{0j}$$

$$\beta'_{1j} = \gamma'_{10} + u'_{1j}$$

$$r'_{ij} \sim N(0, 1)$$

$$\begin{bmatrix} u'_{0j} \\ u'_{1j} \end{bmatrix} \sim N(0, \Omega'_u) \text{ with } \Omega'_u = \begin{bmatrix} \sigma_{u'0}^2 & \\ \sigma_{u'0u'1} & \sigma_{u'1}^2 \end{bmatrix}$$

$$k=30$$

$$n_{jC} = n_{jE} = 30$$

Population model:

$$\frac{Y_{ij}}{\sigma_{r_j}} = \beta'_{0j} + \beta'_{1j} \textit{Treatment}_{ij} + r'_{ij} \quad r'_{ij} \sim N(0, 1)$$

$$\beta'_{0j} = 1.00 + u_{0j} \quad u'_{0j} \sim N(0, 0.5)$$

$$\beta'_{1j} = 1.00 + u_{1j} \quad u'_{1j} \sim N(0, 0.5)$$

Parameter estimates

$$\frac{Y_{ij}}{s_{p_j}} = \beta'_{0j} + \beta'_{1j} \textit{Treatment}_{ij} + r'_{ij} \quad r'_{ij} \sim N(0, 1.00)$$

$$\beta_{0j} = 0.96 + u_{0j} \quad u'_{0j} \sim N(0, 0.42)$$

$$\beta_{1j} = 1.21 + u_{1j} \quad u'_{1j} \sim N(0, 0.49)$$

Study

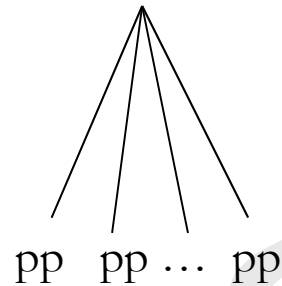
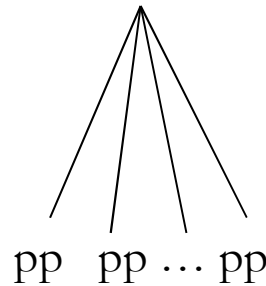
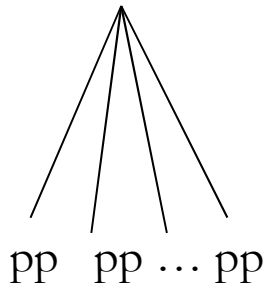
Study 1

Study 2 ...

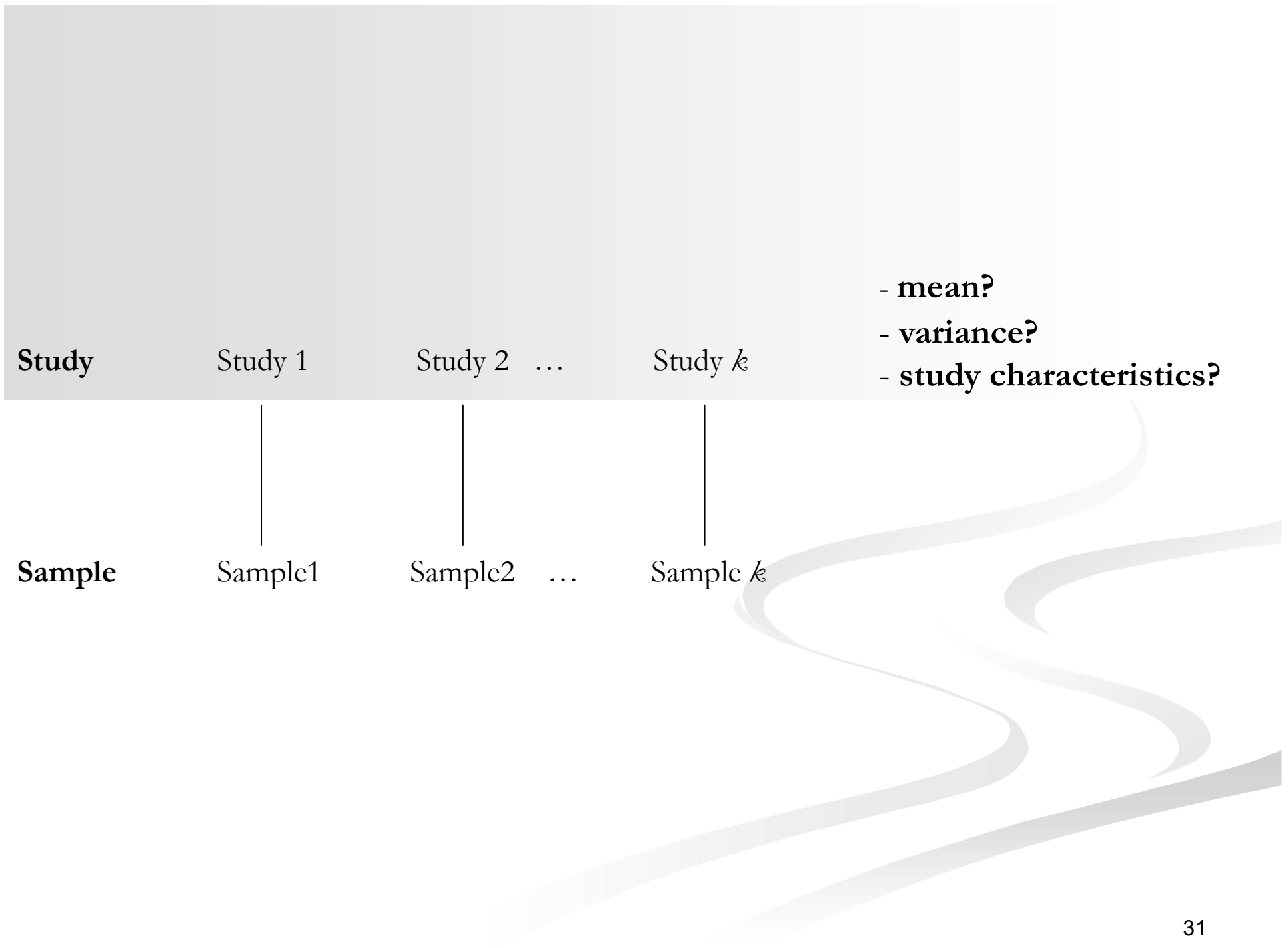
Study k

- mean?
- variance?
- study characteristics?

Participants



- variance?
- Participant characteristics?



$$\frac{\mu_{jE} - \mu_{jC}}{\sigma_{r_j}}$$

σ_{r_j}

Raw data:

$$\frac{Y_{ij}}{\sigma_{r_j}} = \beta'_{0j} + \beta'_{1j} \text{Treatment}_{ij} + r'_{ij}$$

$$r'_{ij} \sim N(0, \sigma_r^2)$$

$$\beta'_{0j} = \gamma'_{00} + u_{0j}$$

$$u'_{0j} \sim N(0, \sigma_{u_0}^2)$$

$$\beta'_{1j} = \gamma'_{10} + u_{1j}$$

$$u'_{1j} \sim N(0, \sigma_{u_0}^2)$$

Effect size data:

$$d_j = \beta'_{1j} + e_j$$

$$e_j \sim N(0, \sigma_{e_j}^2)$$

$$\beta'_{1j} = \gamma'_{10} + u_{1j}$$

$$u'_{1j} \sim N(0, \sigma_{u_0}^2)$$

Raw data:

$$\frac{Y_{ij}}{\sigma_{r_j}} = \beta'_{0j} + \beta'_{1j} \textit{Treatment}_{ij} + r'_{ij} \quad r'_{ij} \sim N(0, 1.00)$$

$$\beta'_{0j} = 0.96 + u_{0j} \quad u'_{0j} \sim N(0, 0.42)$$

$$\beta'_{1j} = 1.21 + u_{1j} \quad u'_{1j} \sim N(0, 0.49)$$

Effect size data:

$$d_j = \beta'_{1j} + e_j \quad e_j \sim N(0, \sigma_{e_j}^2)$$

$$\beta'_{1j} = 1.21 + u_{1j} \quad u'_{1j} \sim N(0, 0.49)$$

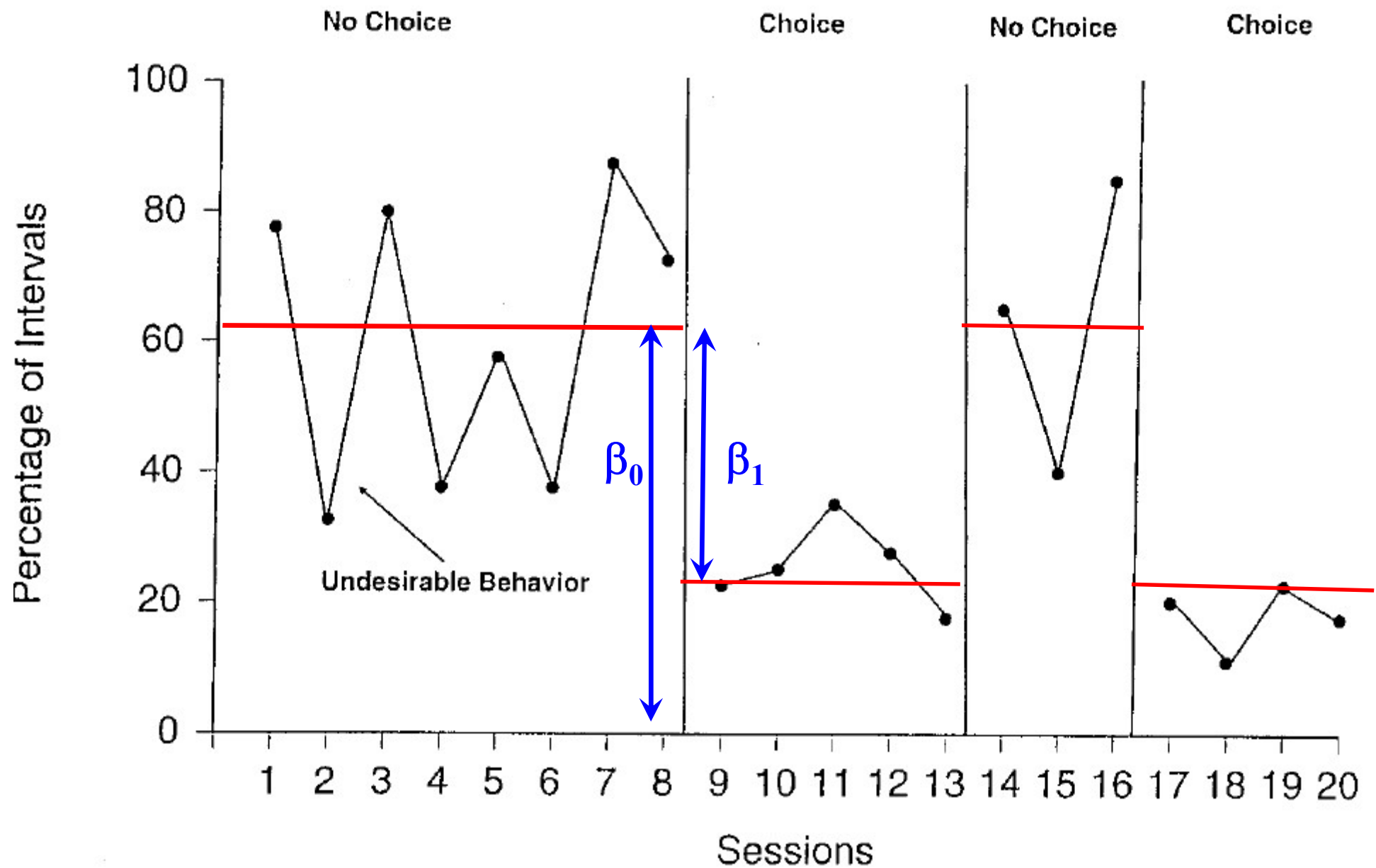
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The MA of Single-Case Experimental studies

Example: A re-analysis of

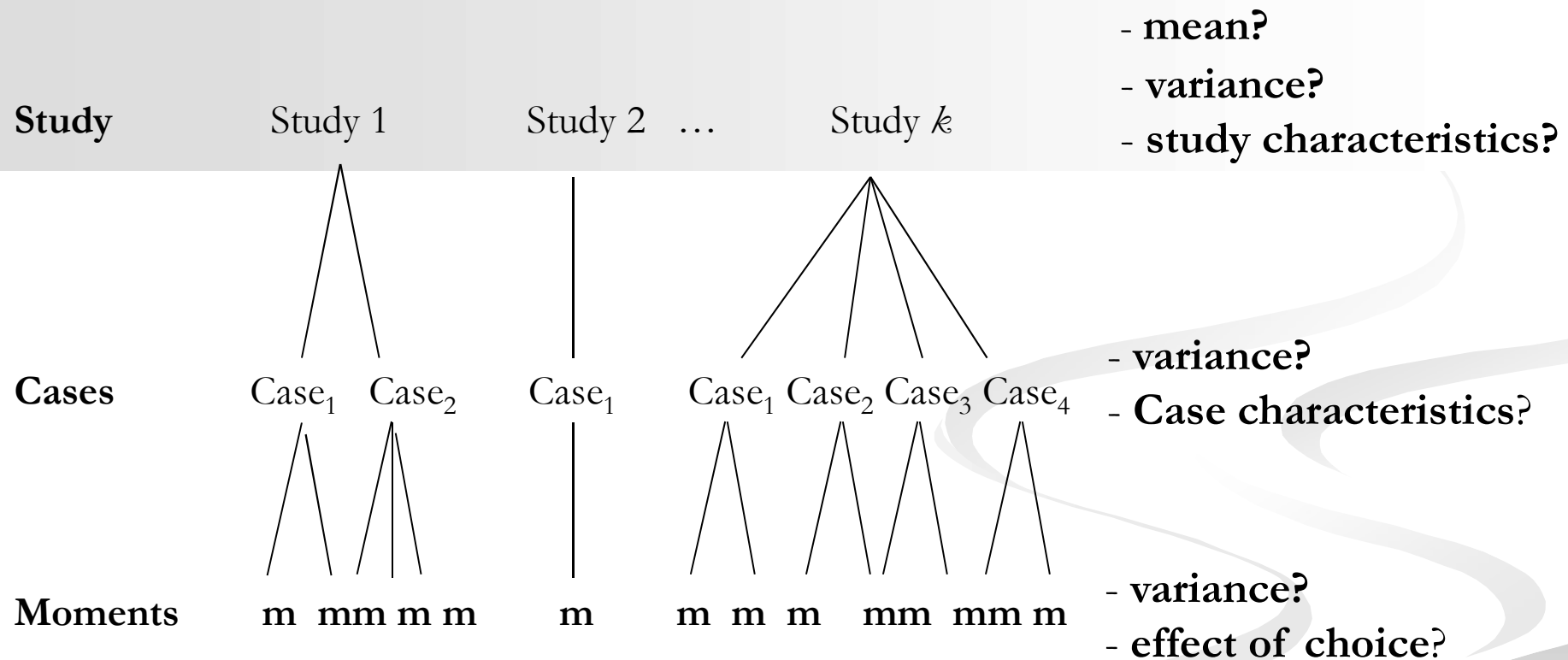
Shogren et al. (2004). The effect of choice-making as an intervention for problem behavior. *Journal of Positive Behavior Interventions*, 6, 228-237.

Van den Noortgate, W., & Onghena, P. (2008). A multilevel meta-analysis of single-subject experimental design studies. *Evidence-Based Communication Assessment and Intervention*, 2, 142-151.



$$Y_i = \beta_0 + \beta_1 \text{Choice}_i + r_i$$

→ 13 single-case experimental studies, 30 cases



A three level model

Level 1:
$$Y_{ijk} = \beta_{0jk} + \beta_{1jk} \textit{Choice}_{ijk} + r_{ijk}$$

Level 2:
$$\beta_{0jk} = \theta_{00k} + u_{0jk}$$

$$\beta_{1jk} = \theta_{10k} + u_{1jk}$$

Level 3:
$$\theta_{00k} = \gamma_{000} + v_{00k}$$

$$\theta_{10k} = \gamma_{100} + v_{10k}$$

Fixed

Intercept	3.07	(0.36)
Choice	-1.84	(0.46)

Variances

Level 3:

Intercept	0	
Choice	0.33	(0.81)
Covariance	0.013	(0.40)

Level 2:

Intercept	2.45	(0.84)
Choice	3.07	(1.21)
Covariance	-2.02	(0.88)

Level 1:

Intercept	1.00	(0.063)
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A three level model

Level 1: $Y_{ijk} = \beta_{0jk} + \beta_{1jk} \textit{Choice}_{ijk} + r_{ijk}$

Level 2: $\beta_{0jk} = \theta_{00k} + \theta_{01} Z_{jk} + u_{0jk}$

$$\beta_{1jk} = \theta_{10k} + \theta_{11} Z_{jk} + u_{1jk}$$

Level 3: $\theta_{00k} = \gamma_{000} + \gamma_{001} W_k + v_{00k}$

$$\theta_{10k} = \gamma_{100} + \gamma_{101} W_k + v_{10k}$$

Fixed

Intercept	3.07	(0.36)	3.34	(1.07)
Choice	-1.84	(0.46)	-2.86	(1.40)
Male			-0.82	(0.76)
Order			0.37	(1.02)
Male x Choice			0.33	(0.91)
Order x Choice			0.98	(1.30)

Variances

Level 3:

Intercept	0	0
Choice	0.33 (0.81)	0.48 (0.92)
Covariance	0.013 (0.40)	-0.15 (0.39)

Level 2:

Intercept	2.45 (0.84)	2.59 (0.93)
Choice	3.07 (1.21)	3.28 (1.35)
Covariance	-2.02 (0.88)	-2.17 (0.99)

Level 1:

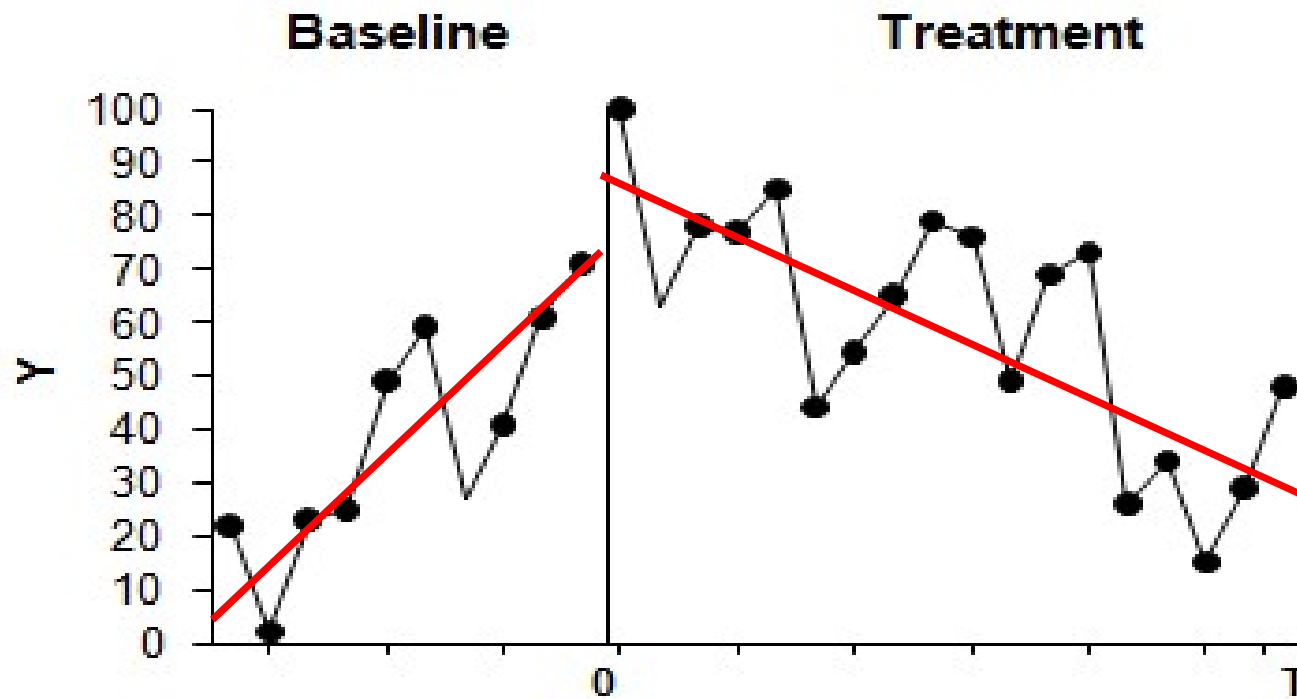
Intercept	1.00 (0.063)	1.00 (0.063) ⁴¹
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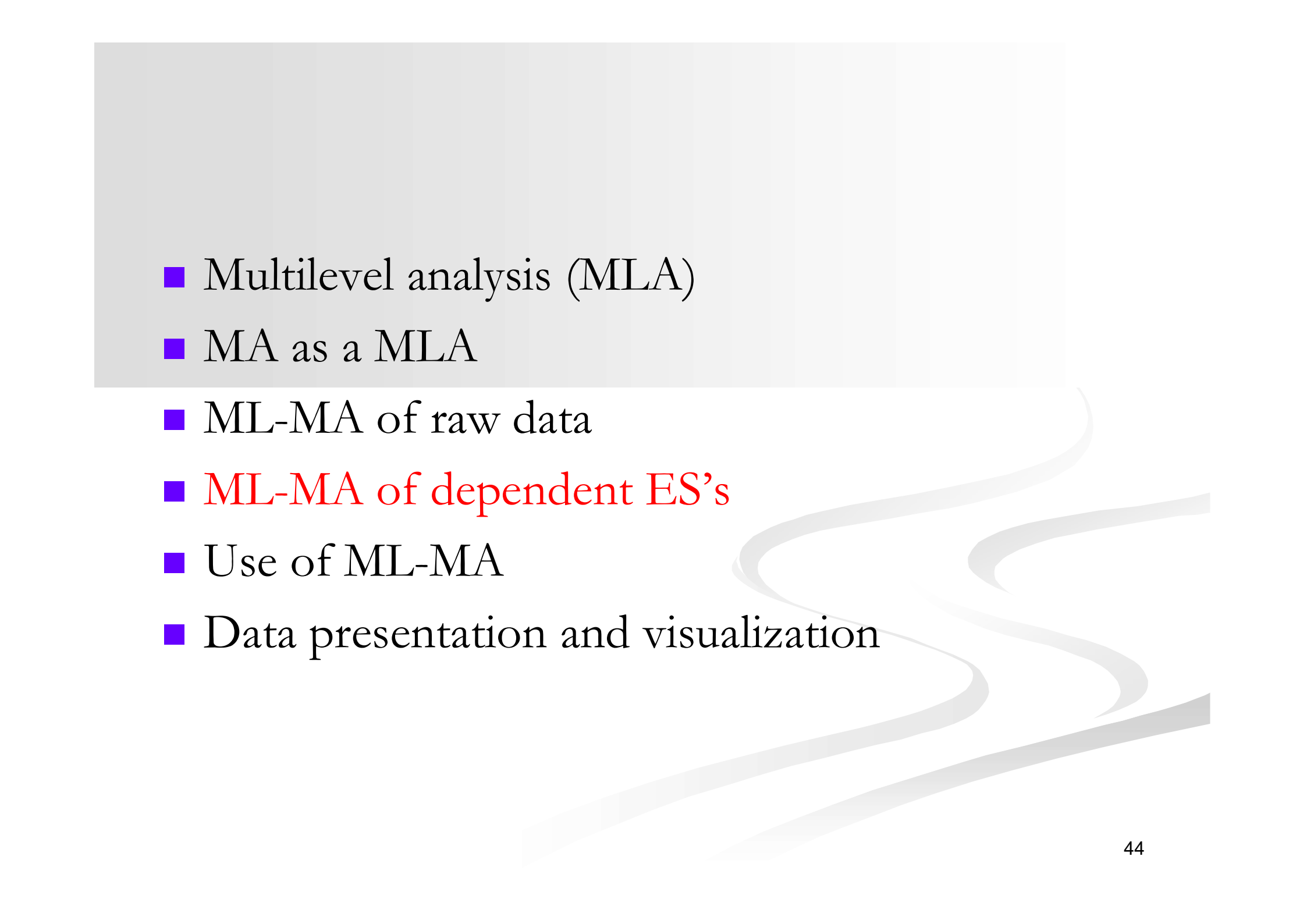
Extensions ...

- Autocorrelation
- Heterogeneity
- Other covariates, e.g. dummy variables for 2 treatments
- Generalized models for count data
- Linear and nonlinear trends
- ...

$$Y_{ij} = \beta_{0j} + \beta_{1j}Time_{ij} + \beta_{2j}Treatment_{ij} + \beta_{3j}Time_{ij} * Treatment_{ij} + r_{ij}$$

(Cfr. Center, Skiba & Casey, 1985-1986)



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Study

θ_1

θ_2

...

θ_k

Sample

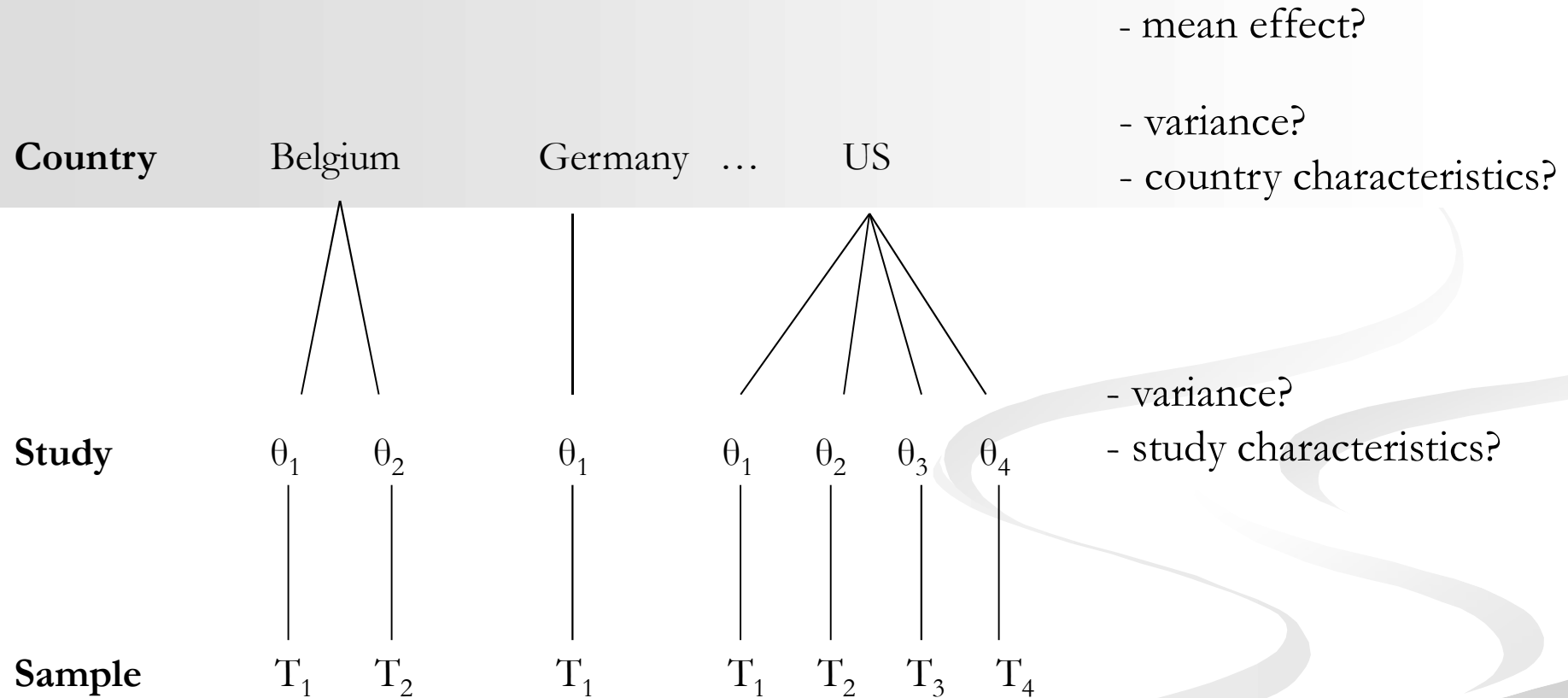
T_1

T_2

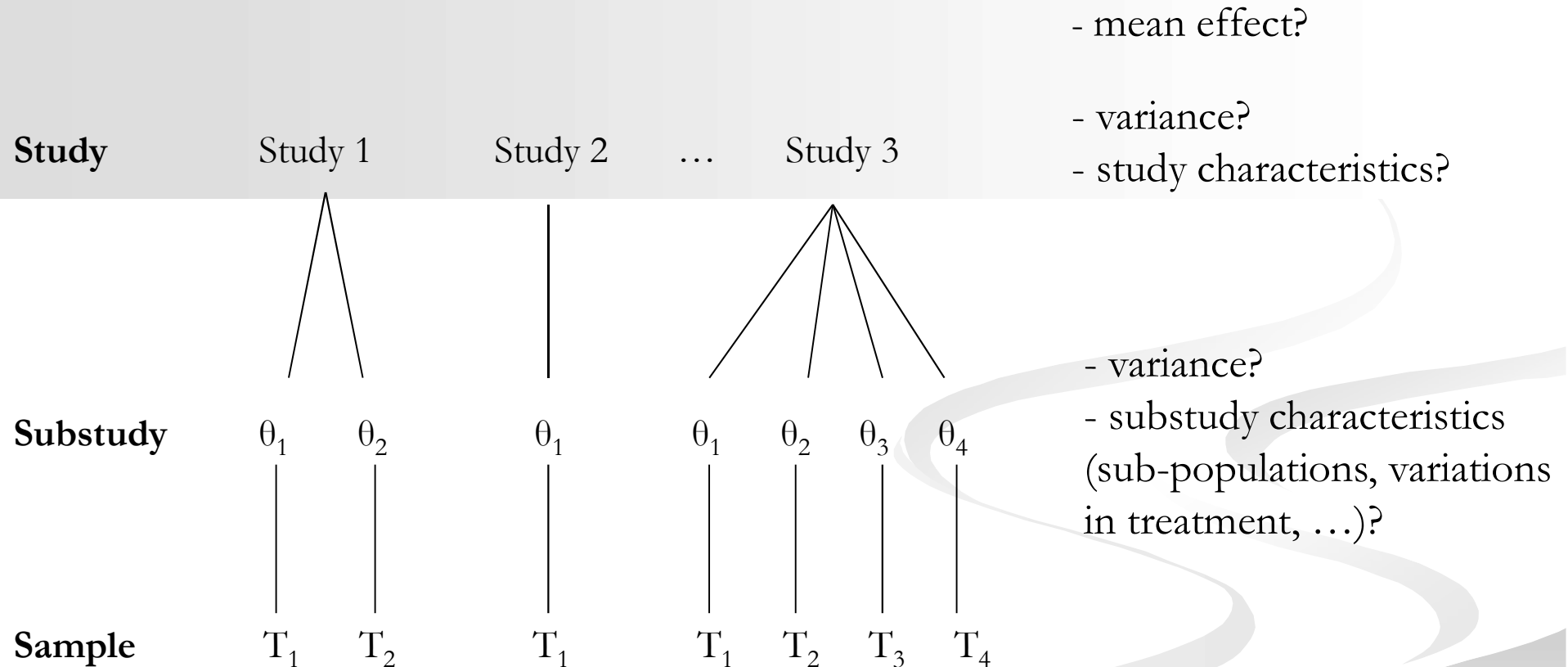
...

T_k

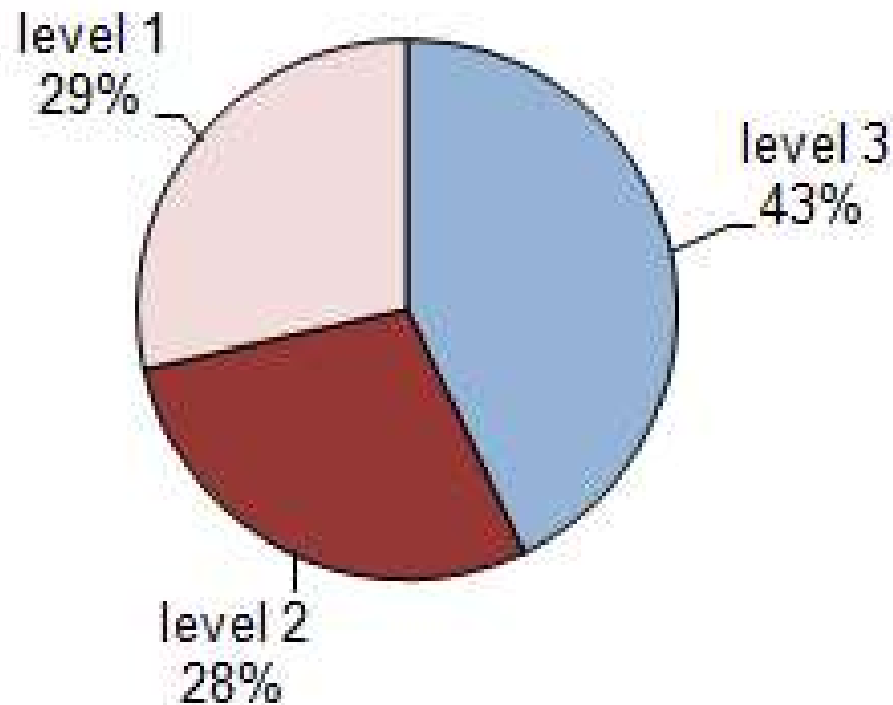
Dependencies over studies



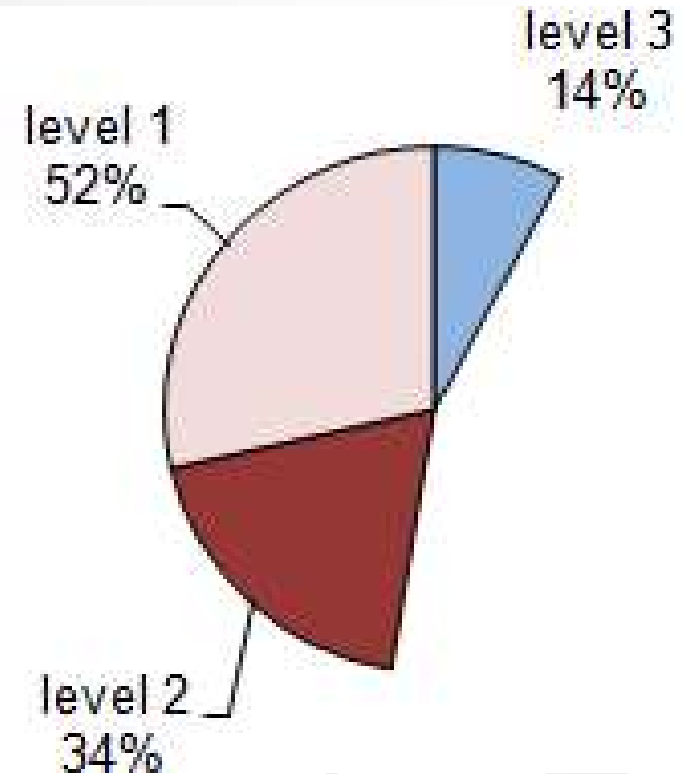
Dependencies within studies



REM



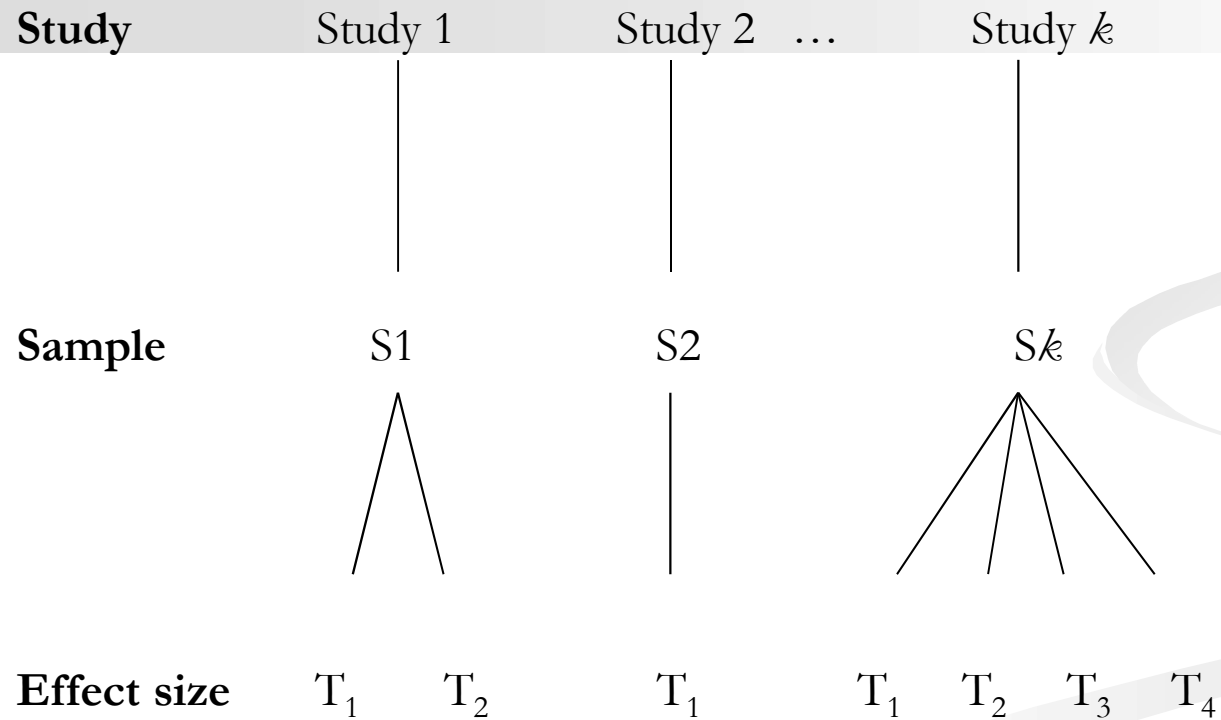
MEM



Van den Bussche, E., Van den Noortgate, W., & Reynvoet, B. (2009). Mechanisms of masked priming: A meta-analysis. *Psychological Bulletin*, 135, 452-477.

(54 studies, 156 experiments)

Dependencies due to multiple outcomes



$\Rightarrow$ Sampling covariance

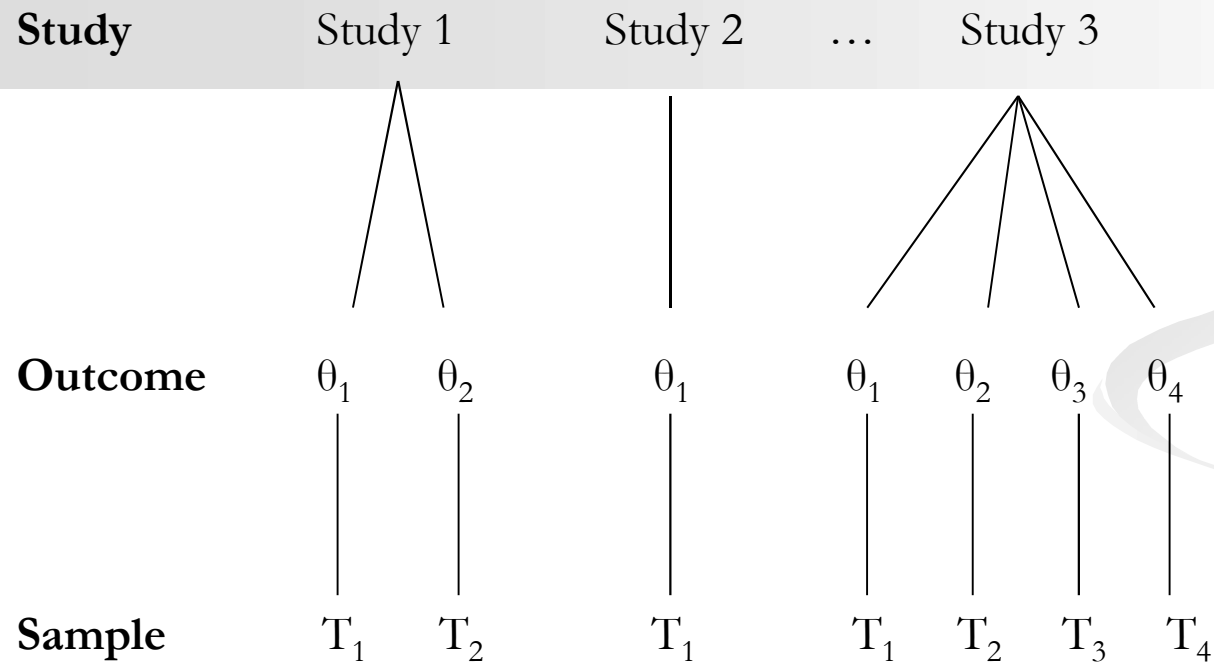
- E.g., multiple SMDs within one study for different outcomes:

$g_1, g_2, g_3, \dots$

$$\sigma_{g_j}^2 \cong \frac{n_E + n_C}{n_E n_C} + \frac{\delta_j^2}{2(n_E + n_C)}$$

$$\sigma_{g_j g_{j'}} \cong \frac{n_E + n_C}{n_E n_C} \rho_{jj'} + \frac{\delta_j \delta_{j'} \rho_{jj'}^2}{2(n_E + n_C)}$$

Can we use this structure?



Parameter		Estimate (SE)	Percentage
Fixed coefficients			
Overall mean	γ_{00}	0.23 (0.026)	
Variance components			
Between-study variance	σ_u^2	0.015	11 %
Between-outcome variance	σ_v^2	0.046	33 %
Median sampling variance	$\sigma_{r_{jk}}^2$	0.078	56 %

(39 studies, 584 effect sizes)

Geeraert, L., Van den Noortgate, W., Grietens, H., & Onghena, P. (2004). The effects of early prevention programs for families with young children at risk for physical child abuse and neglect. A meta-analysis. *Child Maltreatment*, 9, 277-291.

A simulation study

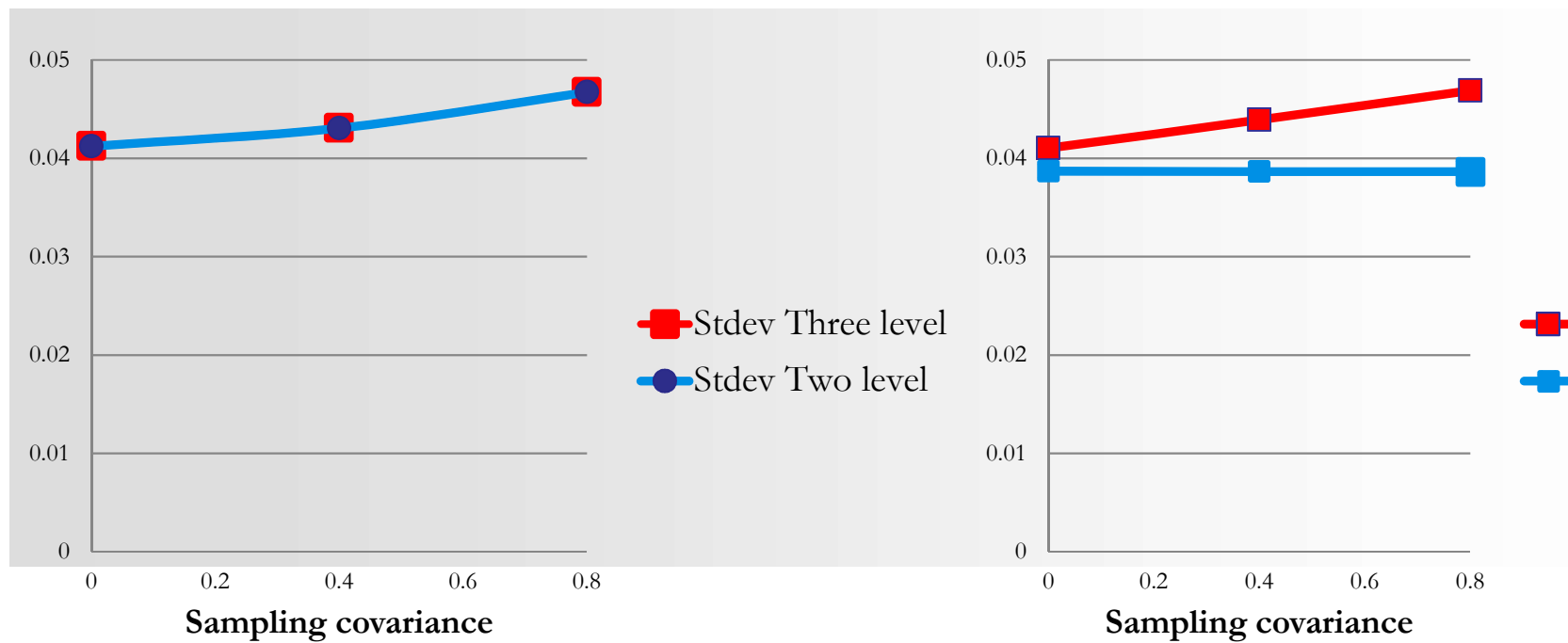
Level 1: participants within studies

$$\begin{cases} Y_{i1k} = \beta_{01k} + \beta_{11k} (Treatment)_{ik} + e_{i1k} \\ Y_{i2k} = \beta_{02k} + \beta_{12k} (Treatment)_{ik} + e_{i2k} \end{cases} \quad \text{with} \quad \begin{bmatrix} e_{i1k} \\ e_{i2k} \end{bmatrix} \sim N(\mathbf{0}, \begin{bmatrix} \sigma_{e_1}^2 & \sigma_{e_2 e_1} \\ \sigma_{e_2 e_1} & \sigma_{e_2}^2 \end{bmatrix})$$

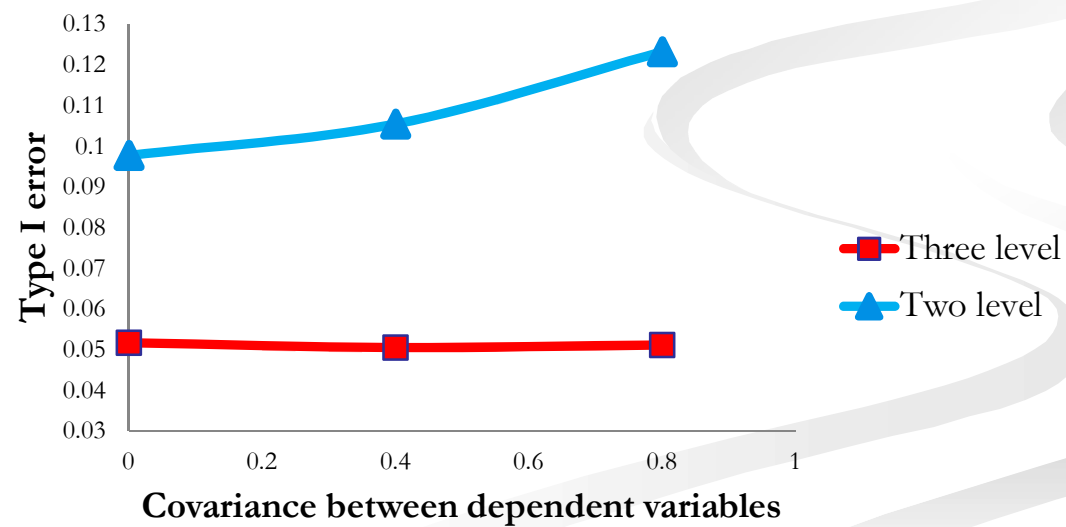
Level 2: (Between) study level

$$\begin{cases} \beta_{01k} = \gamma_{010} + w_{01k} & \text{and} & \beta_{11k} = \gamma_{110} + w_{11k} \\ \beta_{02k} = \gamma_{020} + w_{02k} & \text{and} & \beta_{12k} = \gamma_{120} + w_{12k} \end{cases} \quad \text{with} \quad \mathbf{w}_k \sim N(\mathbf{0}, \Omega_w)$$

Van den Noortgate, W., Lopez-Lopez, J. A., Marin-Martinez, F., & Sánchez-Meca, J. (2013). Three level meta-analyses of dependent effect sizes. *Behavior Research Methods*, 45, 576-594.



Type I errors ($\alpha=.05$)



Interpretation of variances

Level 1: participants within studies

$$\begin{cases} Y_{i1k} = \beta_{01k} + \beta_{11k} (\text{Treatment})_{ik} + e_{i1k} \\ Y_{i2k} = \beta_{02k} + \beta_{12k} (\text{Treatment})_{ik} + e_{i2k} \end{cases} \quad \text{with} \quad \begin{bmatrix} e_{i1k} \\ e_{i2k} \end{bmatrix} \sim N(\mathbf{0}, \begin{bmatrix} \sigma_{e_1}^2 & \sigma_{e_2 e_1} \\ \sigma_{e_2}^2 \end{bmatrix})$$

Level 2: (Between) study level

$$\begin{cases} \beta_{01k} = \gamma_{010} + w_{01k} \\ \beta_{02} = \gamma_{020} + w_{02} \\ \beta_{11k} = \gamma_{110} + w_{11k} \\ \beta_{12k} = \gamma_{120} + w_{12k} \end{cases} \quad \text{with} \quad \mathbf{w}_k \sim N(\mathbf{0}, \begin{bmatrix} \sigma_{w_{01}}^2 & \sigma_{w_{02}w_{01}} & \sigma_{w_{11}w_{01}} & \sigma_{w_{12}w_{01}} \\ \sigma_{w_{02}w_{01}} & \sigma_{w_{02}}^2 & \sigma_{w_{11}w_{02}} & \sigma_{w_{12}w_{02}} \\ \sigma_{w_{11}w_{01}} & \sigma_{w_{11}w_{02}} & \sigma_{w_{11}}^2 & \sigma_{w_{12}w_{11}} \\ \sigma_{w_{12}w_{01}} & \sigma_{w_{12}w_{02}} & \sigma_{w_{12}w_{11}} & \sigma_{w_{12}}^2 \end{bmatrix})$$

Results in between-outcome variance

Results in between-study variance

$$\sigma_w^2 = 0.10$$

$$\gamma_{1j0} = \gamma_{1j'0}$$

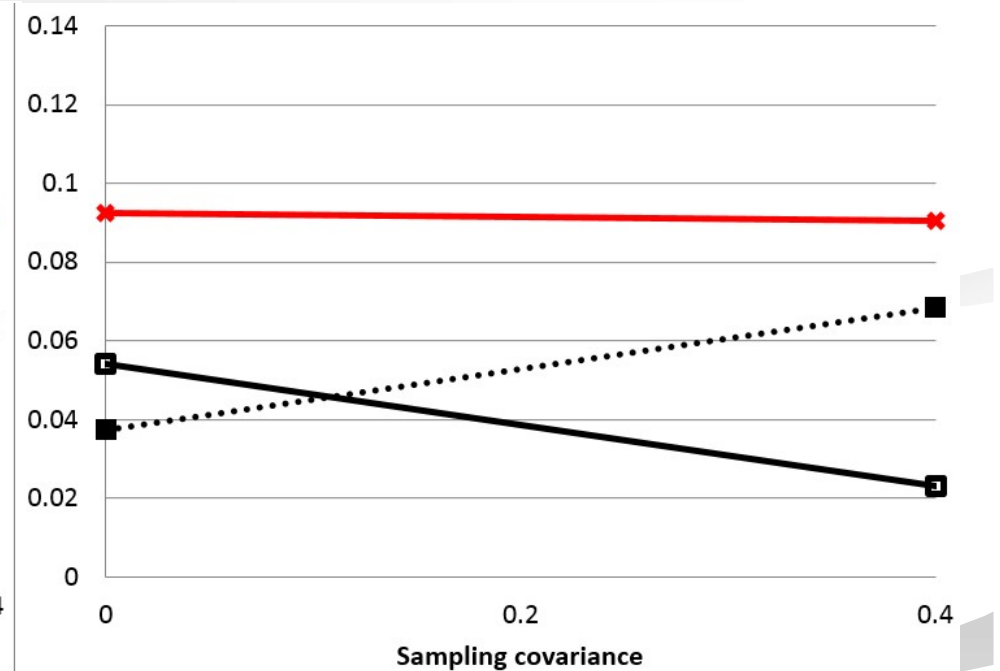
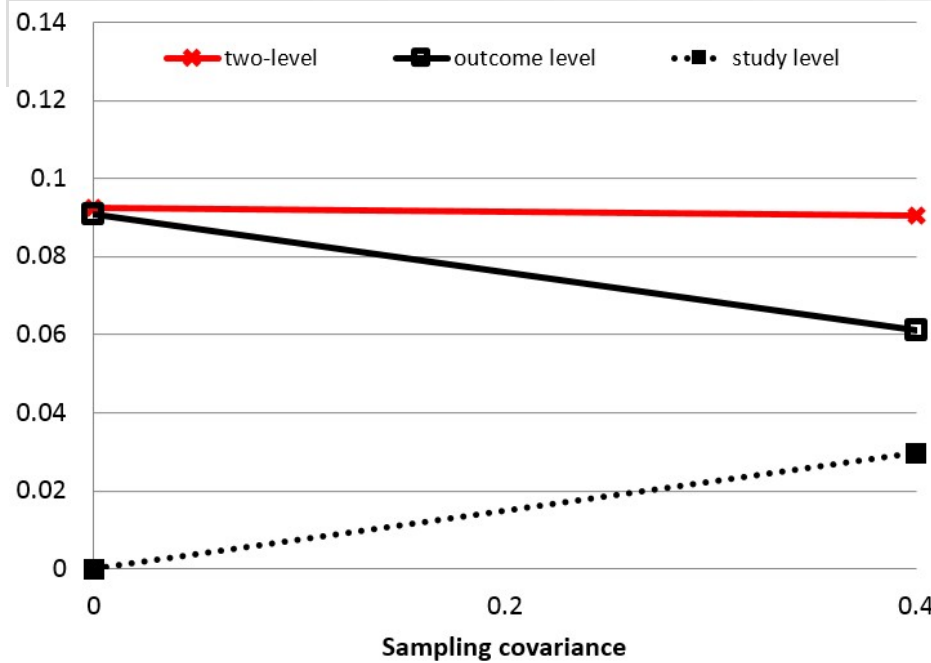
$$\sigma_{w_{1j}w_{1j'}} = 0$$

$$n = 25$$

$$\sigma_w^2 = 0.10$$

$$\gamma_{1j0} = \gamma_{1j'0}$$

$$\sigma_{w_{1j}w_{1j'}} = 0.04$$



$$\text{variance at study level} = \sigma_{w_{1j}w_{1j'}} + \frac{2\sigma_{e_j e_{j'}}}{n}$$

$$\sigma_w^2 = 0.10$$

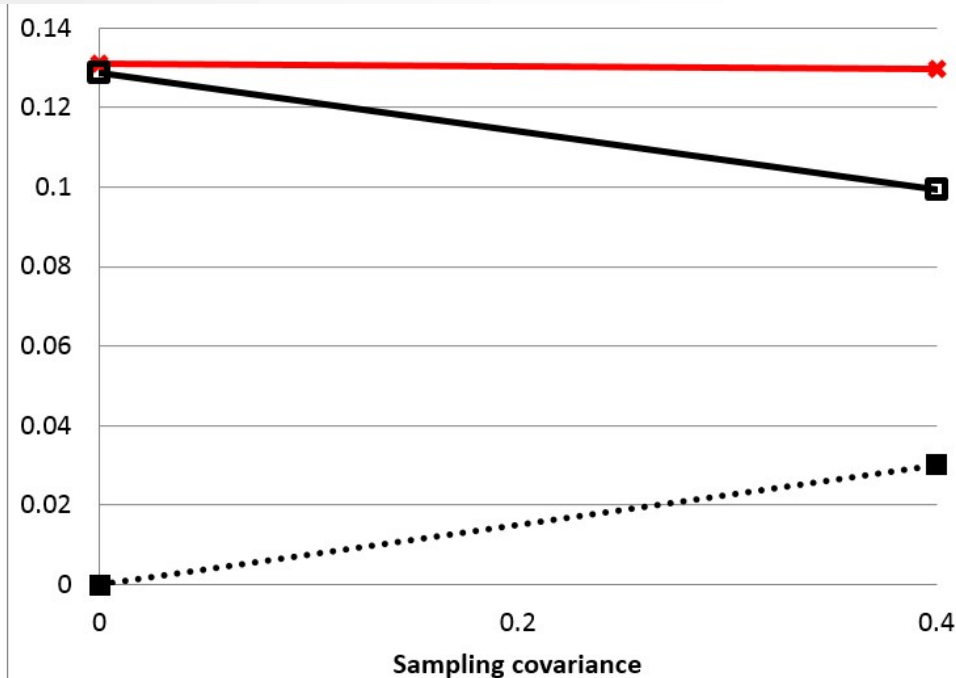
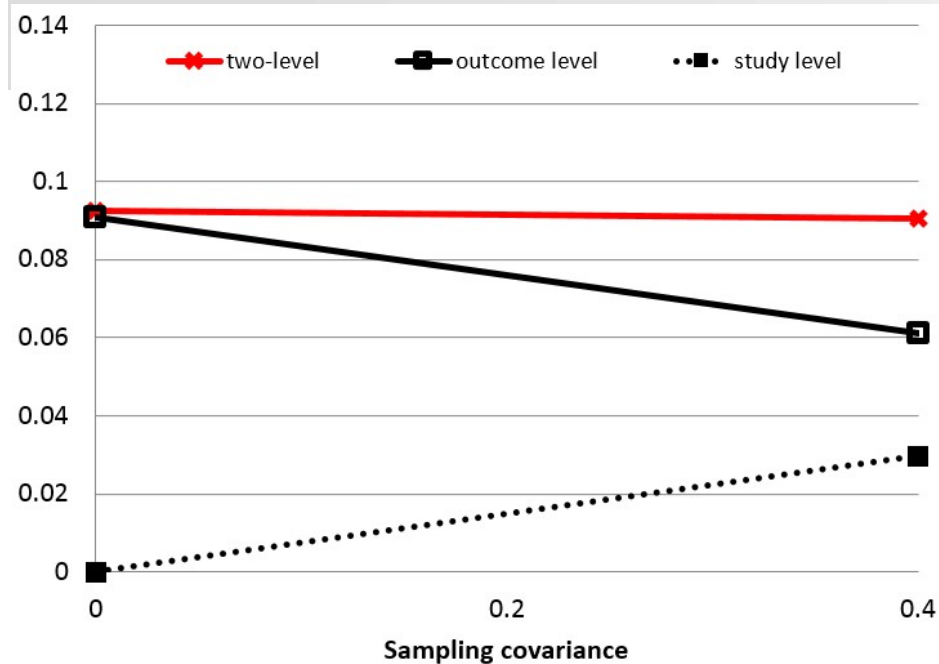
$$\gamma_{1j0} = \gamma_{1j'0}$$

$$\sigma_{w_{1j}w_{1j'}} = 0$$

$$\sigma_w^2 = 0.10$$

$$\sigma_{\gamma_1}^2 = 0.04$$

$$\sigma_{w_{1j}w_{1j'}} = 0$$



Discussion

- 👍 Type I error rate under control
- 👍 Information on two sources of variance
- 👍 Exploring within-study and between-study moderator effects!
- 👎 Complexity & software
- 👎 Assumptions (e.g., same correlation for each pair of outcomes; same between-study variance for each outcome)

Alternatives?

- Ignoring dependence

👍 Simple

👎 Inflated Type I error rate

Alternatives?

- Avoiding dependence
 - Randomly choosing one ES for each study
 - Averaging ES's within a study
 - Performing separate meta-analyses for each kind of treatment or indicator
- 👍 Statistically correct
- 👎 Loss of information (e.g., differences between effect sizes within a study)
- 👎 Correct sampling variance of average depends on the unknown between-outcome correlation!
- 👎 Loss of efficiency (not all data used at the same time)

Alternatives?

- Performing a multivariate meta-analysis, accounting for sampling covariance.

👍 Type I error rate under control

👍 Flexibility

👎 Complexity & software

👎 Often not enough information for estimating ρ

👎 Requires limited set of outcome variables!

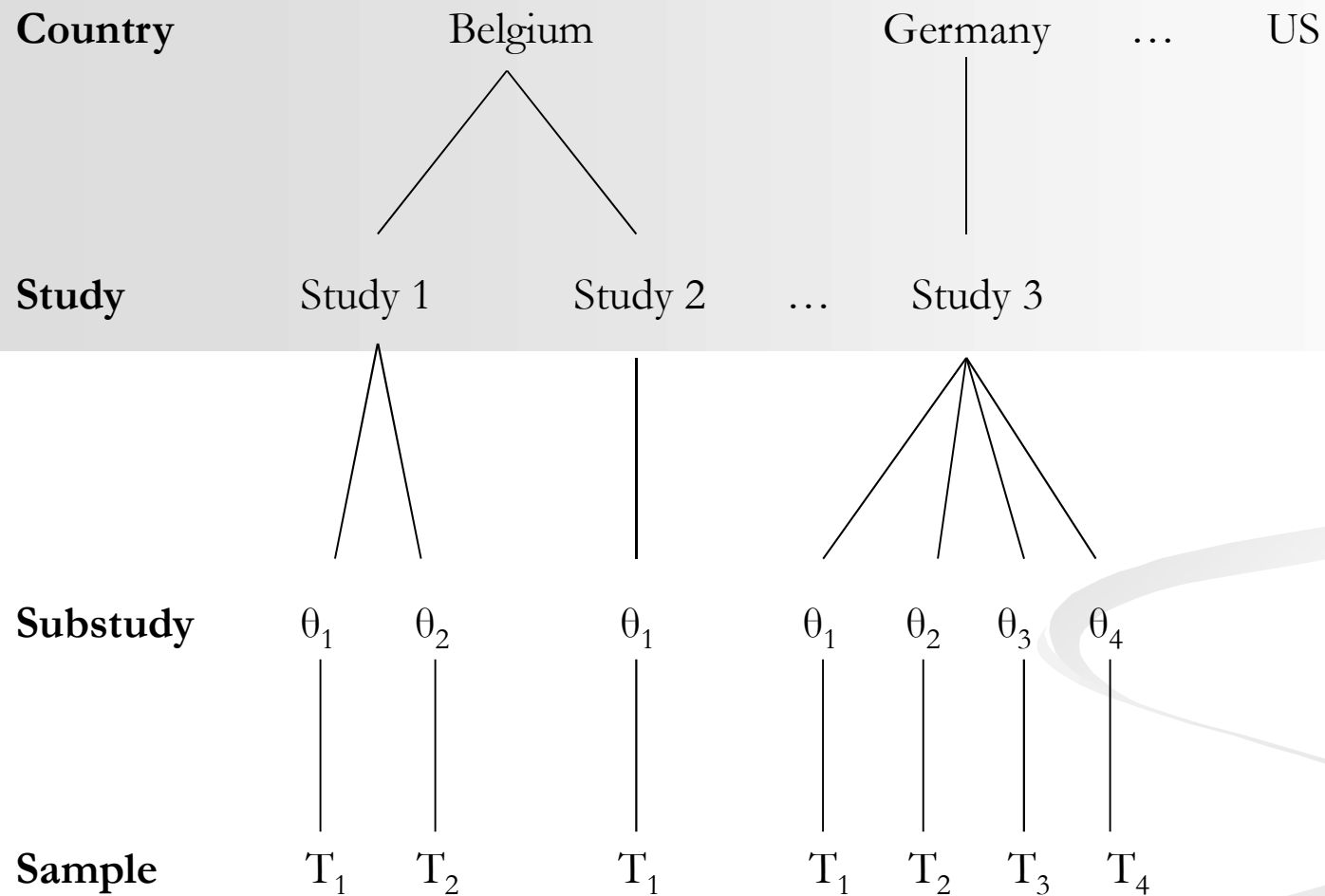
Alternatives?

■ Robust SE estimation (RVE)

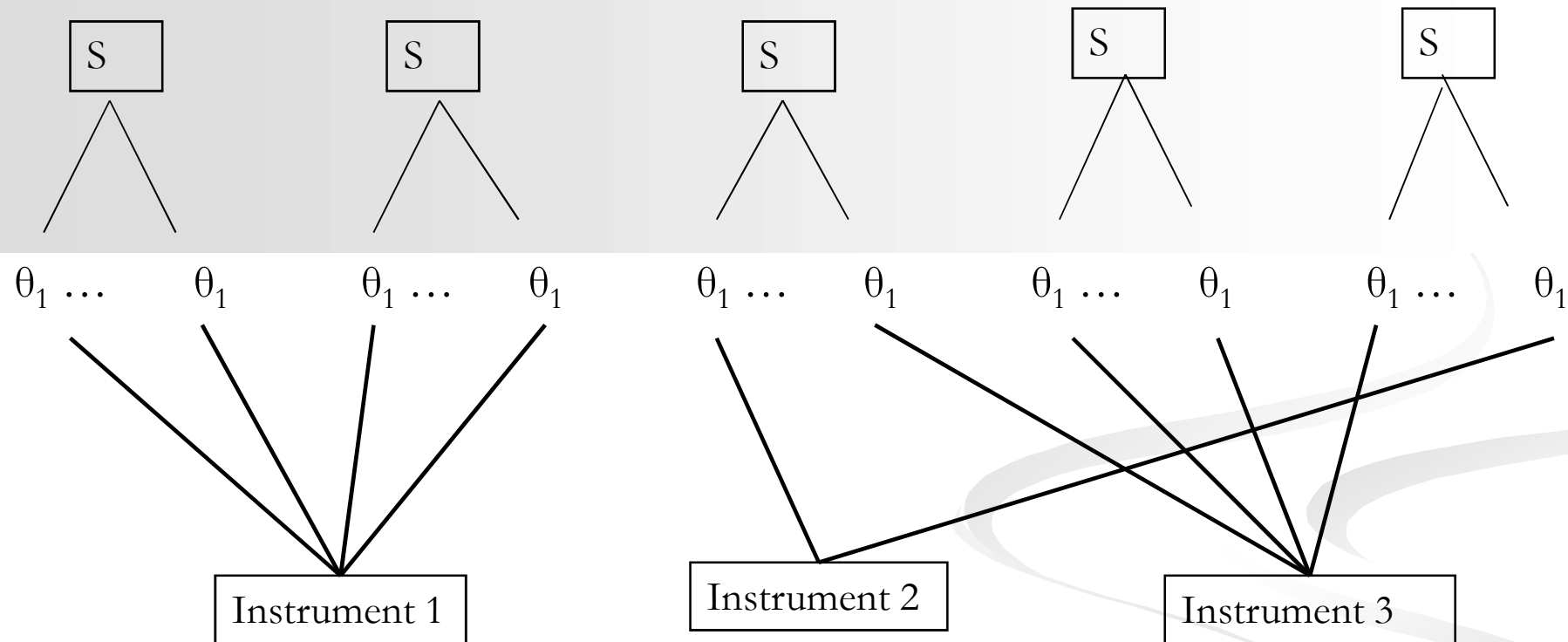
Hedges, L. V., E. Tipton, and M. C. Johnson. 2010. Robust variance estimation in meta-regression with dependent effect size estimates. *Research Synthesis Methods*, 1, 39-65.

- 👍 Type I error rate under control
- 👍 Exploring within- and between-study moderator effects
- 👍 Standard two-level model
- 👎 Complexity & software
- 👎 Assumptions, e.g., same between-study variance for outcomes
- 👎 No separate estimates of between- and within-study variance

(>3)-level meta-analyses ...



Non-hierarchical multilevel meta-analyses ...



Fernández-Castilla, B., et al. (accepted). A demonstration and evaluation of the use of cross-classified random effects models for meta-analysis. *Behavior Research Methods*.

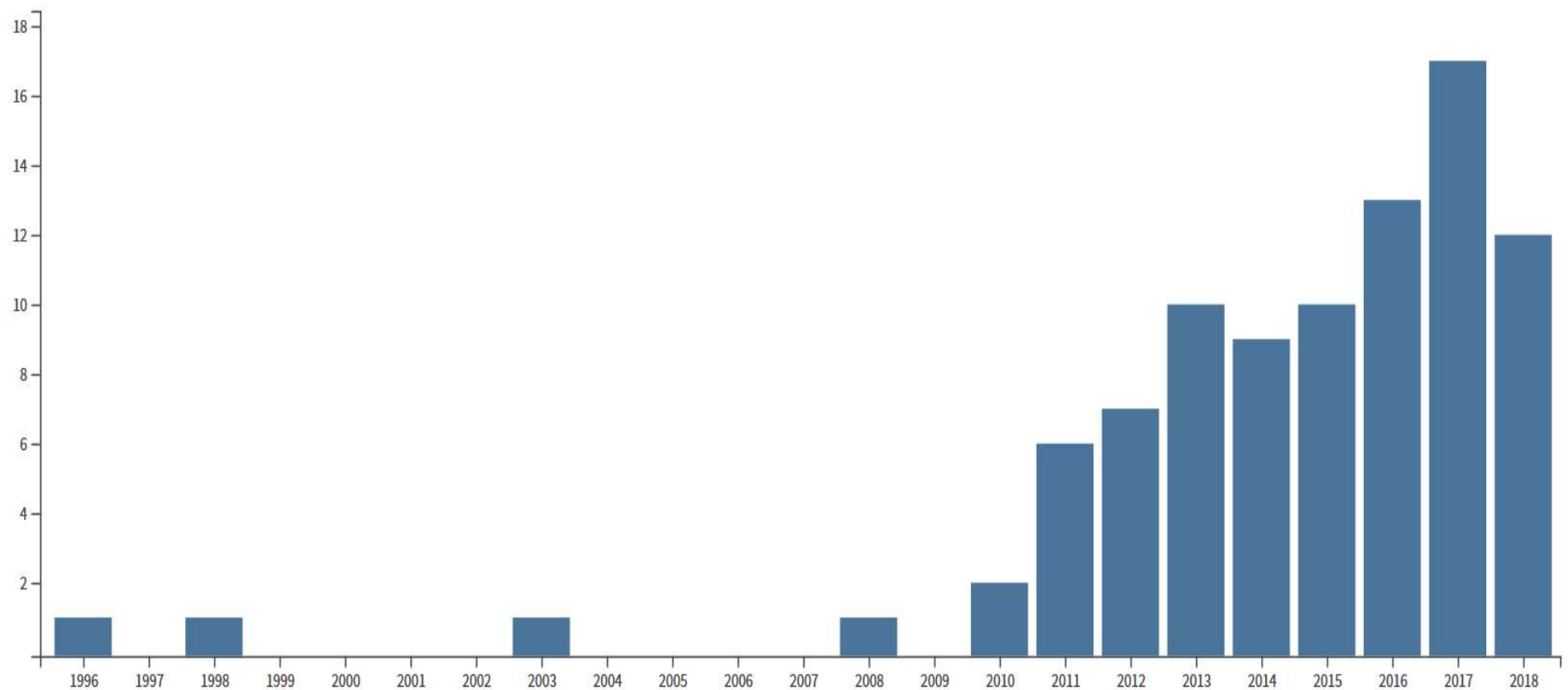
- Multilevel analysis (MLA)
- MA as a MLA
- ML-MA of raw data
- ML-MA of dependent ES's
- The use of ML-MA
- Data presentation and visualization

Search in WOS

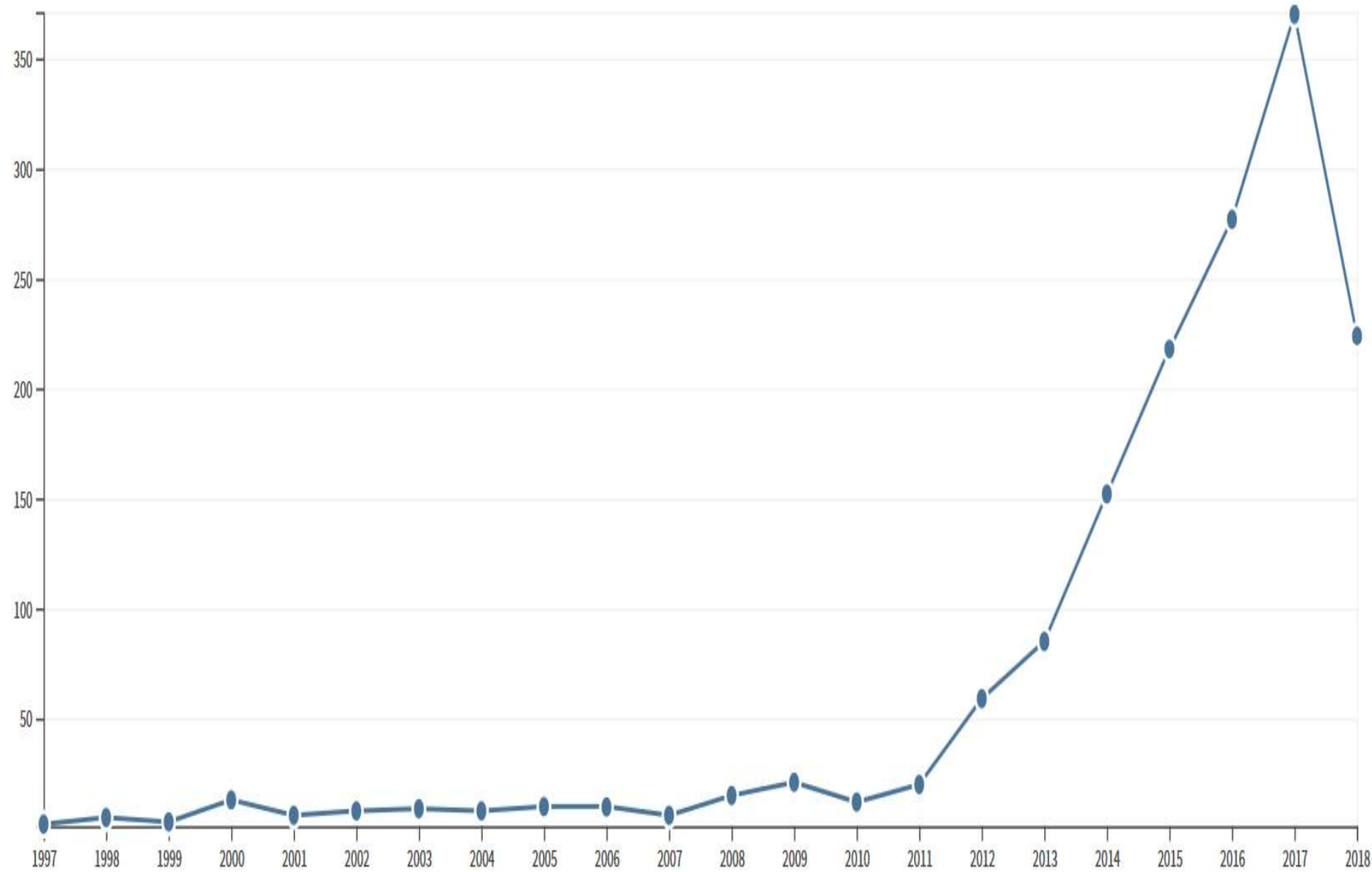
TS=("multilevel meta-analysis" OR "threelevel meta-analysis" OR "three-level meta-analysis")

Total Publications

90



Sum of Times Cited per Year





What applied multilevel meta-analytic papers refer to?

Cheung, M. W. L. (2014). Modeling dependent effect sizes with three-level meta-analyses: A structural equation modeling approach. *Psychological Methods, 19*, 211-229.

Van den Noortgate, W., López-López, J. A., Marín-Martínez, F., & Sánchez-Meca, J. (2013). Three-level meta-analysis of dependent effect sizes. *Behavior Research Methods, 45*, 576-594.

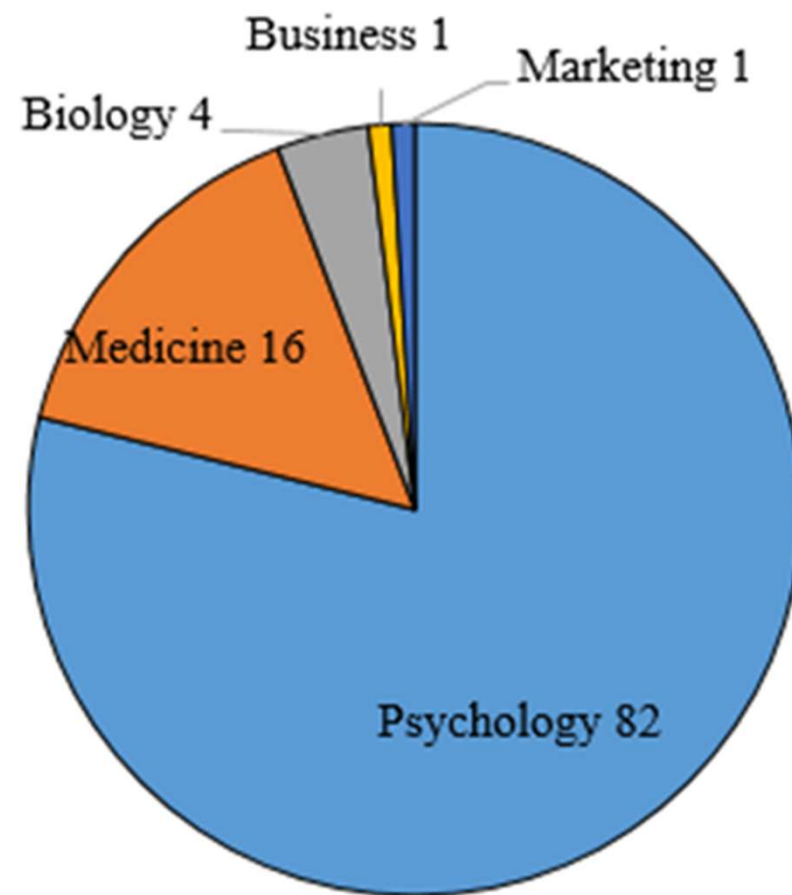
Van den Noortgate, W., López-López, J. A., Marín-Martínez, F., & Sánchez-Meca, J. (2014). Meta-analysis of multiple outcomes: A multilevel approach. *Behavior Research Methods, 47*, 1274-1294.

Results

104 studies found

- 99 studies used 3-level model with intermediate level
- 2 studies used 3-level model with additional upper level (sample – papers – studies)
- 3 studies used a 4-level model
 - Sample – outcomes – studies – countries
 - Sample – outcomes – substudies – studies
 - Sample – subtasks – tasks - studies

Disciplines

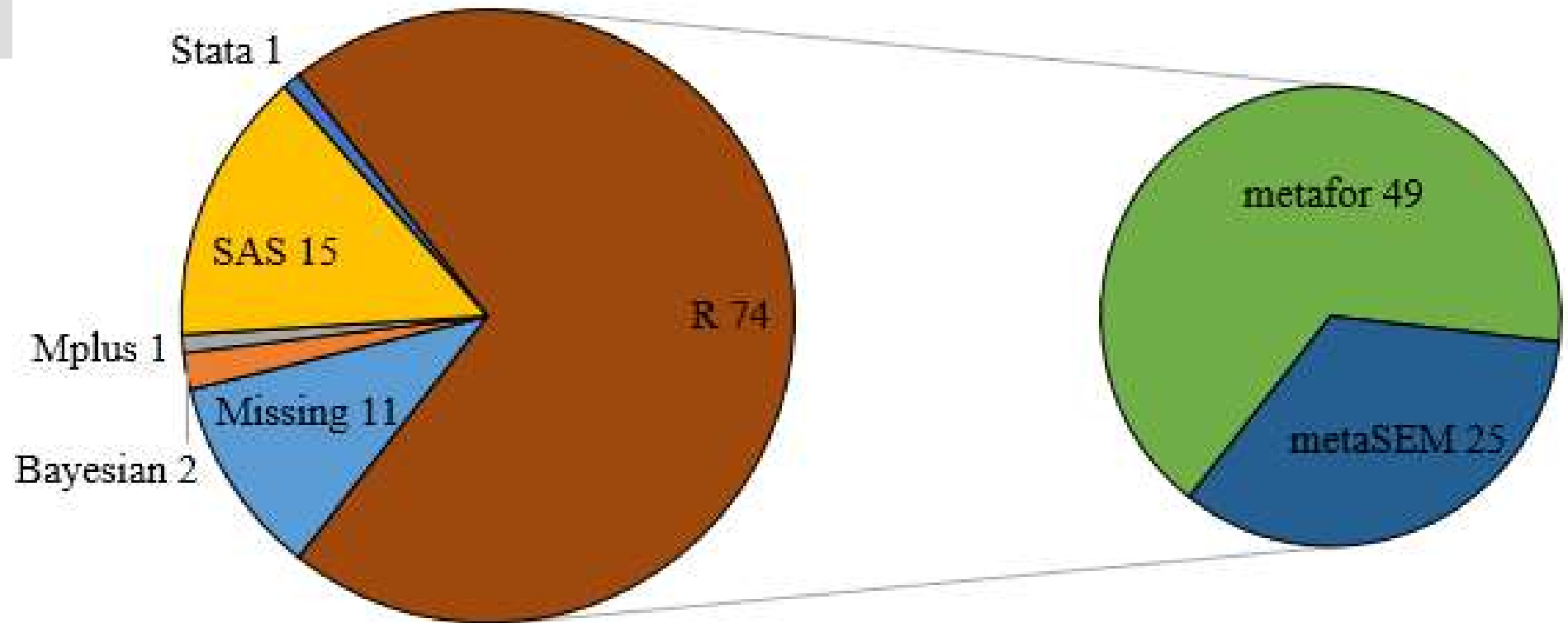


MA of Single case studies

- 178 MAs between 1985-2015
- 22 studies using ML models (19 since 2011)

Jamshidi, L. et al. (under revision) A systematic review of single-case experimental design meta-analyses: Characteristics of study designs, data, and analyses.

Software



- Multilevel analysis (MLA)
- MA as a MLA
- ML-MA of raw data
- ML-MA of dependent ES's
- The use of ML-MA
- Data presentation and visualization

Observed effect sizes

